

$\sqrt{x+y=2}$ Algebra $\pi\theta$ $i\sum f$ Δ CHAPTE 10 TRIGONOMETRIC IDENTITIES

Trigonometric Identities:

Identities involving trigonometric ratios ($\sin \theta, \cos \theta, \text{etc.}$) are called trigonometric identities.

Distance Formula:

Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be two points. If 'd' denotes the distance between them, then

$$d = |PQ| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

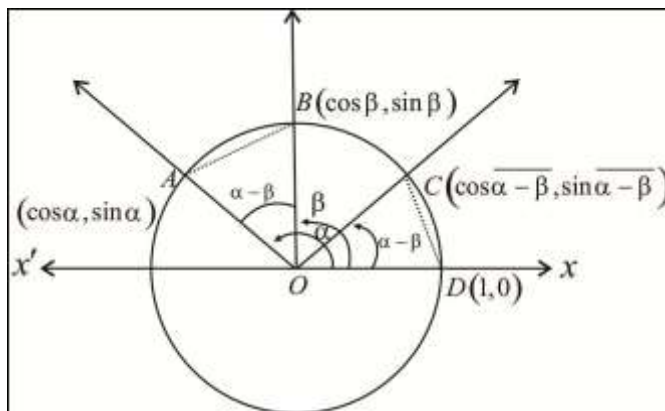
Fundamental law of trigonometry:

Let α and β any two angles (real numbers), then

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

Proof:

Let $\alpha > \beta > 0$, consider a unit circle with centre at origin 'O'.



Let terminal sides of angles α and β cut the unit circle at A and B respectively.

Evidently $\angle AOB = \alpha - \beta$

Take a point 'C' on the unit circle so that $\angle XOC = m\angle AOB = \alpha - \beta$

Join A, B and C, D.

Now angles α , β and $\alpha - \beta$ are in standard position.

$\therefore$ the coordinates of A are $(\cos \alpha, \sin \alpha)$

the coordinates of B are $(\cos \beta, \sin \beta)$

the coordinates of C are $(\cos \alpha - \beta, \sin \alpha - \beta)$

and the coordinates of D are $(1, 0)$

Now $\triangle AOB$ and $\triangle COD$ are congruent by SAS theorem

$$\therefore |AB| = |CD|$$

$$\Rightarrow |AB|^2 = |CD|^2$$

using distance formula

$$(\cos \alpha - \cos \beta)^2 + (\sin \alpha - \sin \beta)^2 = [\cos(\alpha - \beta) - 1]^2 + [\sin(\alpha - \beta) - 0]^2$$

$$\cos^2 \alpha + \cos^2 \beta - 2 \cos \alpha \cos \beta + \sin^2 \alpha + \sin^2 \beta - 2 \sin \alpha \sin \beta$$

$$= \cos^2(\alpha - \beta) + 1 - 2 \cos(\alpha - \beta) + \sin^2(\alpha - \beta)$$

$$(\cos^2 \alpha + \sin^2 \alpha) + (\cos^2 \beta + \sin^2 \beta) - 2 \cos \alpha \cos \beta - 2 \sin \alpha \sin \beta$$

$$= \cos^2(\alpha - \beta) + \sin^2(\alpha - \beta) + 1 - 2 \cos(\alpha - \beta)$$

$$1 + 1 - 2(\cos \alpha \cos \beta + \sin \alpha \sin \beta) = 1 + 1 - 2 \cos(\alpha - \beta)$$

$$2 - 2(\cos \alpha \cos \beta + \sin \alpha \sin \beta) = 2 - 2 \cos(\alpha - \beta)$$

Subtracting "2" from both sides

$$-2(\cos \alpha \cos \beta + \sin \alpha \sin \beta) = -2 \cos(\alpha - \beta) \text{ dividing by '-2' both sides}$$

$$\cos \alpha \cos \beta + \sin \alpha \sin \beta = \cos(\alpha - \beta)$$

$$\text{Hence } \boxed{\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta}$$

Note:

It is true for all values of α and β

Deductions from Fundamental Law:

(1) We know that $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

Putting $\alpha = \frac{\pi}{2}$ in it, we get

$$\cos\left(\frac{\pi}{2} - \beta\right) = \cos \frac{\pi}{2} \cos \beta + \sin \frac{\pi}{2} \sin \beta$$

$$= 0 \cdot \cos \beta + 1 \cdot \sin \beta \quad \left(\because \cos \frac{\pi}{2} = 0, \sin \frac{\pi}{2} = 1 \right)$$

$$\boxed{\therefore \cos\left(\frac{\pi}{2} - \beta\right) = \sin \beta} \quad (i)$$

- (2) We know that $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

Putting $\beta = -\frac{\pi}{2}$ in it, we get

$$\begin{aligned}\cos\left[\alpha - \left(-\frac{\pi}{2}\right)\right] &= \cos \alpha \cos\left(-\frac{\pi}{2}\right) + \sin \alpha \sin\left(-\frac{\pi}{2}\right) \\ \Rightarrow \cos\left(\alpha + \frac{\pi}{2}\right) &= \cos \alpha \cdot 0 + \sin \alpha (-1)\end{aligned}$$

$$\boxed{\therefore \cos\left(\frac{\pi}{2} + \alpha\right) = -\sin \alpha} \quad \text{(ii)}$$

- (3) We know that $\cos\left(\frac{\pi}{2} - \beta\right) = \sin \beta$ [(i) above]

Put $\beta = \frac{\pi}{2} + \alpha$ in it, we get

$$\begin{aligned}\cos\left[\frac{\pi}{2} - \left(\frac{\pi}{2} + \alpha\right)\right] &= \sin\left(\frac{\pi}{2} + \alpha\right) \\ \Rightarrow \cos(-\alpha) &= \sin\left(\frac{\pi}{2} + \alpha\right)\end{aligned}$$

$$\Rightarrow \cos \alpha = \sin\left(\frac{\pi}{2} + \alpha\right) \quad \because \cos(-\alpha) = \cos \alpha$$

$$\boxed{\therefore \sin\left(\frac{\pi}{2} + \alpha\right) = \cos \alpha}$$

- (4) We know that $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

Replace β by $-\beta$

we get

$$\cos[\alpha - (-\beta)] = \cos \alpha \cos(-\beta) + \sin \alpha \sin(-\beta),$$

$$\text{Hence } \boxed{\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta}$$

- (5) We know that $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$

Replace " α " by " $\frac{\pi}{2} + \alpha$ " we get

$$\cos\left[\left(\frac{\pi}{2} + \alpha\right) + \beta\right] = \cos\left(\frac{\pi}{2} + \alpha\right) \cos \beta - \sin\left(\frac{\pi}{2} + \alpha\right) \sin \beta$$

$$\cos\left[\frac{\pi}{2} + (\alpha + \beta)\right] = -\sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos \frac{\pi}{2} \cos(\alpha + \beta) - \sin \frac{\pi}{2} \sin(\alpha + \beta) = -[\sin \alpha \cos \beta + \cos \alpha \sin \beta]$$

$$0 - \sin(\alpha + \beta) = -(\sin \alpha \cos \beta + \cos \alpha \sin \beta)$$

$$-\sin(\alpha + \beta) = -(\sin \alpha \cos \beta + \cos \alpha \sin \beta)$$

multiply both sides by '-1'

$$\text{Hence } \boxed{\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta}$$

(6) We know that $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$

Replace " β " by " $-\beta$ "

$$\sin(\alpha - \beta) = \sin \alpha \cos(-\beta) + \cos \alpha \sin(-\beta)$$

$$\text{Hence } \boxed{\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta}$$

(7) We know that $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

Let $\alpha = 2\pi$, and $\beta = \theta$

$$\therefore \cos(2\pi - \theta) = \cos 2\pi \cos \theta + \sin 2\pi \sin \theta$$

$$= 1 \cdot \cos \theta + 0 \cdot \sin \theta$$

$$= \cos \theta$$

$$\boxed{\cos(2\pi - \theta) = \cos \theta}$$

(8) We know that $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$

Let $\alpha = 2\pi$ and $\beta = \theta$

$$\therefore \sin(2\pi - \theta) = \sin 2\pi \cos \theta - \cos 2\pi \cdot \sin \theta$$

$$= 0 \cdot \cos \theta - 1 \cdot \sin \theta$$

$$= -\sin \theta$$

$$\boxed{\sin(2\pi - \theta) = -\sin \theta}$$

(9) We know that $\tan(\alpha + \beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)}$

$$\tan(\alpha + \beta) = \frac{\sin \alpha \cos \beta + \cos \alpha \sin \beta}{\cos \alpha \cos \beta - \sin \alpha \sin \beta}$$

dividing $\cos \alpha \cos \beta$ in numerator and denominator

$$\begin{aligned}
& \frac{\sin \alpha \cos \beta}{\cos \alpha \cos \beta} + \frac{\cos \alpha \sin \beta}{\cos \alpha \cos \beta} \\
&= \frac{\sin \alpha \cos \beta}{\cos \alpha \cos \beta} - \frac{\sin \alpha \sin \beta}{\cos \alpha \cos \beta} \\
&= \frac{\frac{\sin \alpha}{\cos \alpha} + \frac{\sin \beta}{\cos \beta}}{1 - \frac{\sin \alpha}{\cos \alpha} \cdot \frac{\sin \beta}{\cos \beta}}
\end{aligned}$$

$$\boxed{\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}}$$

(10) We know that $\tan(\alpha - \beta) = \frac{\sin(\alpha - \beta)}{\cos(\alpha - \beta)}$

$$= \frac{\sin \alpha \cos \beta - \cos \alpha \sin \beta}{\cos \alpha \cos \beta + \sin \alpha \sin \beta}$$

Dividing by $\cos \alpha \cos \beta$ in numerator and denominator

$$\begin{aligned}
& \frac{\sin \alpha \cos \beta}{\cos \alpha \cos \beta} - \frac{\cos \alpha \sin \beta}{\cos \alpha \cos \beta} \\
&= \frac{\sin \alpha \cos \beta}{\cos \alpha \cos \beta} + \frac{\sin \alpha \sin \beta}{\cos \alpha \cos \beta} \\
&= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}
\end{aligned}$$

$$\boxed{\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}}$$

Allied Angle:

The angles associated with basic angles of measure θ to a right angle or its multiple are called **allied angles**. So, the angles of measure $90^\circ \pm \theta$, $180^\circ \pm \theta$, $270^\circ \pm \theta$, $360^\circ \pm \theta$, are known as **allied angles**.

$$\begin{cases} \sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta, & \cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta, & \tan\left(\frac{\pi}{2} - \theta\right) = \cot \theta \\ \sin\left(\frac{\pi}{2} + \theta\right) = \cos \theta, & \cos\left(\frac{\pi}{2} + \theta\right) = -\sin \theta, & \tan\left(\frac{\pi}{2} + \theta\right) = -\cot \theta \end{cases}$$

$$\begin{cases} \sin(\pi - \theta) = \sin \theta, & \cos(\pi - \theta) = -\cos \theta, & \tan(\pi - \theta) = -\tan \theta \\ \sin(\pi + \theta) = -\sin \theta, & \cos(\pi + \theta) = -\cos \theta, & \tan(\pi + \theta) = \tan \theta \end{cases}$$

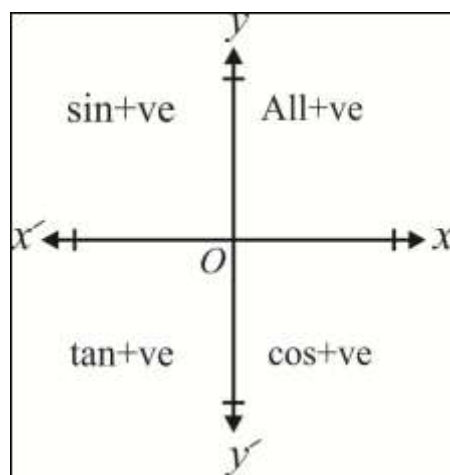
$$\begin{cases} \sin\left(\frac{3\pi}{2}-\theta\right)=-\cos\theta, & \cos\left(\frac{3\pi}{2}-\theta\right)=-\sin\theta, & \tan\left(\frac{3\pi}{2}-\theta\right)=\cot\theta \\ \sin\left(\frac{3\pi}{2}+\theta\right)=-\cos\theta, & \cos\left(\frac{3\pi}{2}+\theta\right)=\sin\theta, & \tan\left(\frac{3\pi}{2}+\theta\right)=-\cot\theta \\ \sin(2\pi-\theta)=-\sin\theta, & \cos(2\pi-\theta)=\cos\theta, & \tan(2\pi-\theta)=-\tan\theta \\ \sin(2\pi+\theta)=\sin\theta, & \cos(2\pi+\theta)=\cos\theta, & \tan(2\pi+\theta)=\tan\theta \end{cases}$$

- (1) If θ is added or subtracted from **odd multiple** of right angle, the trigonometric ratios change into **co-ratios** and vice versa.

i.e. $\sin \longleftrightarrow \cos, \quad \tan \longleftrightarrow \cot, \quad \sec \longleftrightarrow \csc$

- (2) If θ is added to or subtracted from an even multiple of $\frac{\pi}{2}$, the trigonometric ratios shall remain the same.

Measure of the angle	Quad.
$\frac{\pi}{2}-\theta$ or $2\pi+\theta$	I
$\frac{\pi}{2}+\theta$ or $\pi-\theta$	II
$\pi+\theta$ or $\frac{3\pi}{2}-\theta$	III
$\frac{3\pi}{2}+\theta$ or $2\pi-\theta$	IV



Note:

- (1) If basic angle θ is greater than 360° , we subtract multiples of 360° to basic angle

For example:

$$\sin 750^\circ = \sin(750^\circ + (-2)(360^\circ)) = \sin(750^\circ - 720^\circ) = \sin 30^\circ$$

- (2) If basic angle θ is between 0° to 360° we will reduce it into multiples of 90° .

$$\sin 240^\circ = \sin(270^\circ - 30^\circ) = \sin(3 \times 90^\circ - 30^\circ) = -\cos 30^\circ$$

Exercise 10.1

Q.1 Without using tables, find the values of:

(i) $\sin(-780^\circ)$

Solution:

$$\begin{aligned}\sin(-780^\circ) &= \sin(-780^\circ + 2(360^\circ)) \\ &= \sin(-780^\circ + 720^\circ) \\ &= \sin(-60^\circ) \\ &= -\sin 60^\circ \\ &= -\frac{\sqrt{3}}{2}\end{aligned}$$

(ii) $\cot(-855^\circ)$

Solution:

$$\begin{aligned}\cot(-855^\circ) &= \cot(-855^\circ + 2(360^\circ)) \\ &= \cot(-855^\circ + 720^\circ) \\ &= \cot(-135^\circ) \\ &= -\cot 135^\circ \\ &= -\cot(90^\circ + 45^\circ) \\ &= -(-\tan 45^\circ) \\ &= 1\end{aligned}$$

(iii) $\operatorname{cosec}(2040^\circ)$

Solution:

$$\begin{aligned}\text{Here } \operatorname{cosec}(2040^\circ) &= \operatorname{cosec}(2040^\circ + (-5)(360^\circ)) \\ &= \operatorname{cosec}(2040^\circ - 1800^\circ) \\ &= \operatorname{cosec}(240^\circ) \\ &= \operatorname{cosec}(180^\circ + 60^\circ) \\ &= -\operatorname{cosec} 60^\circ\end{aligned}$$

$$= -\frac{2}{\sqrt{3}}$$

(iv) $\sec(-960^\circ)$

Solution:

$$\begin{aligned}\sec(-960^\circ) &= \sec(-960^\circ + 3(360^\circ)) \\ &= \sec(-960^\circ + 1080^\circ) \\ &= \sec 120^\circ \\ &= \sec(90^\circ + 30^\circ) \\ &= -\operatorname{cosec} 30^\circ \\ &= -2\end{aligned}$$

(v) $\tan(1110^\circ)$

Solution:

$$\begin{aligned}\text{Here } \tan(1110^\circ) &= \tan(1110^\circ + (-3)(360^\circ)) \\ &= \tan(1110^\circ - 1080^\circ) \\ &= \tan 30^\circ \\ &= \frac{1}{\sqrt{3}}\end{aligned}$$

(vi) $\sin(-300^\circ)$

Solution:

$$\begin{aligned}\text{Here } \sin(-300^\circ) &= \sin(-300^\circ + 360^\circ) \\ &= \sin 60^\circ \\ &= \frac{\sqrt{3}}{2}\end{aligned}$$

Q.2 Express each of the following as a trigonometric function of an angle of positive degree measure of less than 45° .

(i) $\sin(196^\circ)$

Solution:

$$\begin{aligned}\sin 196^\circ &= \sin(180^\circ + 16^\circ) \\ &= \sin(90^\circ \times 2 + 16^\circ) \\ &= -\sin 16^\circ\end{aligned}$$

(ii) $\cos 147^\circ$

Solution:

$$\begin{aligned}\cos 147^\circ &= \cos(180^\circ - 33^\circ) \\ &= \cos(2 \times 90^\circ - 33^\circ) \\ &= -\cos 33^\circ\end{aligned}$$

(iii) $\sin 319^\circ$

Solution:

$$\begin{aligned}\sin 319^\circ &= \sin(360^\circ - 41^\circ) \\ &= \sin(4 \times 90^\circ - 41^\circ) \\ &= -\sin 41^\circ\end{aligned}$$

(iv) $\cos 254^\circ$

Solution:

$$\begin{aligned}\cos 254^\circ &= \cos(270^\circ - 16^\circ) \\ &= \cos(3 \times 90^\circ - 16^\circ) \\ &= -\sin 16^\circ\end{aligned}$$

(v) $\tan 294^\circ$

Solution:

$$\begin{aligned}\tan 294^\circ &= \tan(270^\circ + 24^\circ)\end{aligned}$$

$$= \tan(3 \times 90^\circ + 24^\circ)$$

$$= -\cot 24^\circ$$

(vi) $\cos 728^\circ$

Solution:

$$\begin{aligned}\cos 728^\circ &= \cos(728^\circ + (-2)(360^\circ)) \\ &= \cos(728^\circ - 720^\circ) \\ &= \cos 8^\circ\end{aligned}$$

(vii) $\sin(-625^\circ)$

Solution:

$$\begin{aligned}\sin(-625^\circ) &= \sin(-625^\circ + 2(360^\circ)) \\ &= \sin(-625^\circ + 720^\circ) \\ &= \sin 95^\circ \\ &= \sin(90^\circ + 5^\circ) \\ &= \cos 5^\circ\end{aligned}$$

(viii) $\cos(-435^\circ)$

Solution:

$$\begin{aligned}\cos(-435^\circ) &= \cos(-435^\circ + 360^\circ) \\ &= \cos(-75^\circ) \\ &= \cos 75^\circ \\ &= \cos(90^\circ - 15^\circ) \\ &= \sin 15^\circ\end{aligned}$$

(ix) $\sin 150^\circ$

Solution:

$$\begin{aligned}\sin 150^\circ &= \sin(180^\circ - 30^\circ) \\ &= \sin(2 \times 90^\circ - 30^\circ) \\ &= \sin 30^\circ\end{aligned}$$

Q.3 Prove the following:

(i) $\sin(180^\circ + \alpha)\sin(90^\circ - \alpha) = -\sin \alpha \cos \alpha$

Solution:

$$\begin{aligned}\text{L.H.S} &= \sin(180^\circ + \alpha)\sin(90^\circ - \alpha) \\ &= (-\sin \alpha)(\cos \alpha) \\ &= -\sin \alpha \cos \alpha \\ &= \text{R.H.S}\end{aligned}$$

(ii) $\sin 780^\circ \sin 480^\circ + \cos 120^\circ \sin 30^\circ = \frac{1}{2}$

Solution:

$$\begin{aligned}\text{L.H.S} &= \sin 780^\circ \sin 480^\circ + \cos 120^\circ \sin 30^\circ \\ &= \sin(780^\circ + (-2)(360^\circ))\sin(480^\circ + (-1)(360^\circ)) + \cos(90^\circ + 30^\circ)\sin 30^\circ \\ &= \sin(780^\circ - 720^\circ)\sin(480^\circ - 360^\circ) + (-\sin 30^\circ)\sin 30^\circ \\ &= \sin 60^\circ \sin 120^\circ - \frac{1}{2} \cdot \frac{1}{2} \\ &= \frac{\sqrt{3}}{2} \cdot \sin(90^\circ + 30^\circ) - \frac{1}{4} \\ &= \frac{\sqrt{3}}{2} \cdot (\cos 30^\circ) - \frac{1}{4} \\ &= \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} - \frac{1}{4} \\ &= \frac{3}{4} - \frac{1}{4} \\ &= \frac{3-1}{4} \\ &= \frac{2}{4} \\ &= \frac{1}{2} = \text{R.H.S}\end{aligned}$$

(iii) $\cos 306^\circ + \cos 234^\circ + \cos 162^\circ + \cos 18^\circ = 0$

Solution:

$$\begin{aligned}\text{L.H.S} &= \cos 306^\circ + \cos 234^\circ + \cos 162^\circ + \cos 18^\circ \\ &= \cos(360^\circ - 54^\circ) + \cos(180^\circ + 54^\circ) + \cos(180^\circ - 18^\circ) + \cos 18^\circ \\ &= \cos(4 \times 90^\circ - 54^\circ) + \cos(2 \times 90^\circ + 54^\circ) + \cos(2 \times 90^\circ - 18^\circ) + \cos 18^\circ \\ &= \cos 54^\circ - \cos 54^\circ - \cos 18^\circ + \cos 18^\circ \\ &= 0 = \text{R.H.S}\end{aligned}$$

$$(iv) \quad \cos 330^\circ \sin 600^\circ + \cos 120^\circ \sin 150^\circ = 1$$

Solution:

$$\begin{aligned} \text{L.H.S} &= \cos 330^\circ \sin 600^\circ + \cos 120^\circ \sin 150^\circ \\ &= \cos(270^\circ + 60^\circ) \sin(600^\circ + (-2)(360^\circ)) + \cos(90^\circ + 30^\circ) \sin(90^\circ + 60^\circ) \\ &= \cos(3 \times 90^\circ + 60^\circ) \sin(600^\circ - 720^\circ) + (-\sin 30^\circ)(\cos 60^\circ) \\ &= \sin 60^\circ \sin(-120^\circ) - \sin 30^\circ \cos 60^\circ \\ &= -\sin 60^\circ \sin(90^\circ + 30^\circ) - \frac{1}{2} \cdot \frac{1}{2} \\ &= -\frac{\sqrt{3}}{2} \cdot \cos 30^\circ - \frac{1}{4} \\ &= -\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} - \frac{1}{4} \\ &= \frac{-3}{4} - \frac{1}{4} \\ &= \frac{-4}{4} \\ &= -1 = \text{R.H.S} \end{aligned}$$

Q.4 Prove that:

$$(i) \quad \frac{\sin^2(\pi + \theta) \tan\left(\frac{3\pi}{2} + \theta\right)}{\cot^2\left(\frac{3\pi}{2} - \theta\right) \cos^2(\pi - \theta) \operatorname{cosec}(2\pi - \theta)} = \cos \theta$$

Solution:

$$\begin{aligned} \text{L.H.S} &= \frac{\sin^2(\pi + \theta) \tan\left(\frac{3\pi}{2} + \theta\right)}{\cot^2\left(\frac{3\pi}{2} - \theta\right) \cos^2(\pi - \theta) \operatorname{cosec}(2\pi - \theta)} \\ &= \frac{[\sin(\pi + \theta)]^2 \tan\left(\frac{3\pi}{2} + \theta\right)}{\left[\cot\left(\frac{3\pi}{2} - \theta\right)\right]^2 [\cos(\pi - \theta)]^2 \cdot \operatorname{cosec}(2\pi - \theta)} \\ &= \frac{(-\sin \theta)^2 \cdot (-\cot \theta)}{(\tan \theta)^2 (-\cos \theta)^2 \cdot (-\operatorname{cosec} \theta)} \end{aligned}$$

$$\begin{aligned}
 &= \frac{\sin^2 \theta (-\cot \theta)}{\tan^2 \theta \cdot \cos^2 \theta (-\operatorname{cosec} \theta)} \\
 &= \frac{(\sin^2 \theta) \left(-\frac{\cos \theta}{\sin \theta} \right)}{\left(\frac{\sin^2 \theta}{\cos^2 \theta} \right) \cdot \cos^2 \theta \cdot \left(-\frac{1}{\sin \theta} \right)} \\
 &= \frac{-\sin \theta \cos \theta}{-\sin \theta} \\
 &= \cos \theta = \text{R.H.S}
 \end{aligned}$$

$$(ii) \quad \frac{\cos(90^\circ + \theta) \sec(-\theta) \tan(180^\circ - \theta)}{\sec(360^\circ - \theta) \sin(180^\circ + \theta) \cot(90^\circ - \theta)} = -1$$

Solution:

$$\begin{aligned}
 \text{L.H.S} &= \frac{\cos(90^\circ + \theta) \sec(-\theta) \tan(180^\circ - \theta)}{\sec(360^\circ - \theta) \sin(180^\circ + \theta) \cot(90^\circ - \theta)} \\
 &= \frac{(-\sin \theta)(\sec \theta)(-\tan \theta)}{(\sec \theta)(-\sin \theta)(\tan \theta)} \\
 &= -1 = \text{R.H.S}
 \end{aligned}$$

Q.5 If α, β, γ are the angles of a triangle ABC , then prove that

$$(i) \quad \sin(\alpha + \beta) = \sin \gamma$$

Solution:

Since α, β, γ are the angles of triangle ABC

$$\text{So } \alpha + \beta + \gamma = 180^\circ$$

$$\Rightarrow \alpha + \beta = 180^\circ - \gamma$$

$$\Rightarrow \sin(\alpha + \beta) = \sin(180^\circ - \gamma)$$

$$\Rightarrow \sin(\alpha + \beta) = \sin \gamma$$

$$(ii) \quad \cos\left(\frac{\alpha + \beta}{2}\right) = \sin \frac{\gamma}{2}$$

Solution:

$$\text{Here } \alpha + \beta + \gamma = 180^\circ$$

$$\Rightarrow \alpha + \beta = 180^\circ - \gamma$$

$$\Rightarrow \frac{\alpha + \beta}{2} = \frac{180^\circ - \gamma}{2}$$

$$\Rightarrow \frac{\alpha + \beta}{2} = 90^\circ - \frac{\gamma}{2}$$

$$\Rightarrow \cos\left(\frac{\alpha + \beta}{2}\right) = \cos\left(90^\circ - \frac{\gamma}{2}\right)$$

$$\Rightarrow \cos\left(\frac{\alpha + \beta}{2}\right) = \sin \frac{\gamma}{2}$$

$$(iii) \quad \cos(\alpha + \beta) = -\cos \gamma$$

Solution:

$$\text{Here } \alpha + \beta + \gamma = 180^\circ$$

$$\Rightarrow \alpha + \beta = 180^\circ - \gamma$$

$$\Rightarrow \cos(\alpha + \beta) = \cos(180^\circ - \gamma)$$

$$\Rightarrow \cos(\alpha + \beta) = -\cos \gamma$$

$$(iv) \quad \tan(\alpha + \beta) + \tan \gamma = 0$$

Solution:

$$\text{Here } \alpha + \beta + \gamma = 180^\circ$$

$$\Rightarrow \alpha + \beta = 180^\circ - \gamma$$

$$\Rightarrow \tan(\alpha + \beta) = \tan(180^\circ - \gamma)$$

$$\Rightarrow \tan(\alpha + \beta) = -\tan \gamma$$

$$\Rightarrow \tan(\alpha + \beta) + \tan \gamma = 0$$

Exercise 10.2**Q.1 Prove that**

(i) $\sin(180^\circ + \theta) = -\sin \theta$

Solution:

$$\begin{aligned}
 \text{L.H.S.} &= \sin(180^\circ + \theta) \\
 &= \sin 180^\circ \cos \theta + \cos 180^\circ \sin \theta \\
 &= (0)(\cos \theta) + (-1)(\sin \theta) \\
 &= -\sin \theta = \text{R.H.S}
 \end{aligned}$$

(ii) $\cos(180^\circ + \theta) = -\cos \theta$

Solution:

$$\begin{aligned}
 \text{L.H.S} &= \cos(180^\circ + \theta) \\
 &= \cos 180^\circ \cos \theta - \sin 180^\circ \sin \theta \\
 &= (-1)(\cos \theta) - (0)(\sin \theta) \\
 &= -\cos \theta = \text{R.H.S}
 \end{aligned}$$

(iii) $\tan(270^\circ - \theta) = \cot \theta$

Solution:

$$\begin{aligned}
 \text{L.H.S} &= \tan(270^\circ - \theta) \\
 &= \frac{\sin(270^\circ - \theta)}{\cos(270^\circ - \theta)} \\
 &= \frac{(\sin 270^\circ)(\cos \theta) - (\cos 270^\circ)(\sin \theta)}{(\cos 270^\circ)(\cos \theta) + (\sin 270^\circ)(\sin \theta)} \\
 &= \frac{(-1) \cdot \cos \theta - 0 \cdot \sin \theta}{0 \cdot \cos \theta + (-1) \cdot \sin \theta} \\
 &= \frac{(-1)(\cos \theta)}{(-1)(\sin \theta)} \\
 &= \frac{-\cos \theta}{-\sin \theta} \\
 &= \frac{\cos \theta}{\sin \theta} \\
 &= \cot \theta = \text{R.H.S}
 \end{aligned}$$

(iv) $\cos(\theta - 180^\circ) = -\cos \theta$

Solution:

$$\begin{aligned}
 \text{L.H.S} &= \cos(\theta - 180^\circ) \\
 &= \cos \theta \cos(180^\circ) + \sin \theta \sin(180^\circ) \\
 &= (\cos \theta)(-1) + (\sin \theta)(0) \\
 &= -\cos \theta = \text{R.H.S}
 \end{aligned}$$

(v) $\cos(270^\circ + \theta) = \sin \theta$

Solution:

$$\begin{aligned}
 \text{L.H.S} &= \cos(270^\circ + \theta) \\
 &= (\cos 270^\circ)(\cos \theta) - (\sin 270^\circ)(\sin \theta) \\
 &= 0 - (-1)\sin \theta \\
 &= \sin \theta = \text{R.H.S}
 \end{aligned}$$

(vi) $\sin(\theta + 270^\circ) = -\cos \theta$

Solution:

$$\begin{aligned}
 \text{L.H.S} &= \sin(\theta + 270^\circ) \\
 &= (\sin \theta)(\cos 270^\circ) + (\cos \theta)(\sin 270^\circ) \\
 &= (\sin \theta)(0) + (\cos \theta)(-1) \\
 &= -\cos \theta = \text{R.H.S}
 \end{aligned}$$

(vii) $\tan(180^\circ + \theta) = \tan \theta$

Solution:

$$\begin{aligned}
 \text{L.H.S} &= \tan(180^\circ + \theta) \\
 &= \frac{\tan 180^\circ + \tan \theta}{1 - \tan 180^\circ \tan \theta} \\
 &= \frac{0 + \tan \theta}{1 - (0) \tan \theta} \\
 &= \tan \theta = \text{R.H.S}
 \end{aligned}$$

(viii) $\cos(360^\circ - \theta) = \cos \theta$

Solution:

$$\begin{aligned}
 \text{L.H.S} &= \cos(360^\circ - \theta) \\
 &= \cos 360^\circ \cdot \cos \theta + \sin 360^\circ \cdot \sin \theta \\
 &= (1)(\cos \theta) + (0)(\sin \theta) \\
 &= \cos \theta = \text{R.H.S}
 \end{aligned}$$

Q.2 Find the values of the following:

(i) $\sin 15^\circ$

Solution:

$$\begin{aligned}\sin 15^\circ &= \sin(45^\circ - 30^\circ) \\ &= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ \\ &= \left(\frac{1}{\sqrt{2}}\right) \times \left(\frac{\sqrt{3}}{2}\right) - \left(\frac{1}{\sqrt{2}}\right) \times \left(\frac{1}{2}\right) \\ &= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} \\ &= \frac{\sqrt{3}-1}{2\sqrt{2}}\end{aligned}$$

(ii) $\cos 15^\circ$

Solution:

$$\begin{aligned}\cos 15^\circ &= \cos(45^\circ - 30^\circ) \\ &= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ \\ &= \left(\frac{1}{\sqrt{2}}\right) \times \left(\frac{\sqrt{3}}{2}\right) + \left(\frac{1}{\sqrt{2}}\right) \times \left(\frac{1}{2}\right) \\ &= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} \\ &= \frac{\sqrt{3}+1}{2\sqrt{2}}\end{aligned}$$

(iii) $\tan 15^\circ$

Solution:

$$\begin{aligned}\tan 15^\circ &= \tan(45^\circ - 30^\circ) \\ &= \frac{\tan 45^\circ - \tan 30^\circ}{1 + (\tan 45^\circ)(\tan 30^\circ)} \\ &= \frac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} \\ &= \frac{\frac{\sqrt{3}-1}{\sqrt{3}}}{\frac{\sqrt{3}+1}{\sqrt{3}}} \\ &= \frac{\sqrt{3}-1}{\sqrt{3}+1}\end{aligned}$$

(iv) $\sin 105^\circ$

Solution:

$$\begin{aligned}\sin 105^\circ &= \sin(60^\circ + 45^\circ) \\ &= \sin 60^\circ \cos 45^\circ + \cos 60^\circ \sin 45^\circ \\ &= \left(\frac{\sqrt{3}}{2}\right) \times \left(\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{2}\right) \times \left(\frac{1}{\sqrt{2}}\right) \\ &= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} \\ &= \frac{\sqrt{3}+1}{2\sqrt{2}}\end{aligned}$$

(v) $\cos 105^\circ$

Solution:

$$\begin{aligned}\cos 105^\circ &= \cos(60^\circ + 45^\circ) \\ &= \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ \\ &= \left(\frac{1}{2}\right) \times \left(\frac{1}{\sqrt{2}}\right) - \left(\frac{\sqrt{3}}{2}\right) \times \left(\frac{1}{\sqrt{2}}\right) \\ &= \frac{1}{2\sqrt{2}} - \frac{\sqrt{3}}{2\sqrt{2}} \\ &= \frac{1-\sqrt{3}}{2\sqrt{2}}\end{aligned}$$

(vi) $\tan 105^\circ$

Solution:

$$\begin{aligned}\tan 105^\circ &= \tan(60^\circ + 45^\circ) \\ &= \frac{\tan 60^\circ + \tan 45^\circ}{1 - \tan 60^\circ \tan 45^\circ} \\ &= \frac{\sqrt{3}+1}{1-\sqrt{3}}\end{aligned}$$

Q.3 Prove that

$$(i) \quad \sin(45^\circ + \alpha) = \frac{1}{\sqrt{2}}(\sin \alpha + \cos \alpha)$$

Solution:

$$\begin{aligned} \text{L.H.S} &= \sin(45^\circ + \alpha) \\ &= \sin 45^\circ \cos \alpha + \cos 45^\circ \sin \alpha \\ &= \frac{1}{\sqrt{2}} \cos \alpha + \frac{1}{\sqrt{2}} \sin \alpha \\ &= \frac{1}{\sqrt{2}}(\sin \alpha + \cos \alpha) = \text{R.H.S} \end{aligned}$$

$$(ii) \quad \cos(\alpha + 45^\circ) = \frac{1}{\sqrt{2}}(\cos \alpha - \sin \alpha)$$

Solution:

$$\begin{aligned} \text{L.H.S} &= \cos(\alpha + 45^\circ) \\ &= \cos \alpha \cos 45^\circ - \sin \alpha \sin 45^\circ \\ &= (\cos \alpha) \left(\frac{1}{\sqrt{2}} \right) - (\sin \alpha) \left(\frac{1}{\sqrt{2}} \right) \\ &= \frac{1}{\sqrt{2}}(\cos \alpha - \sin \alpha) = \text{R.H.S} \end{aligned}$$

Q.4 Prove that

$$(i) \quad \tan(45^\circ + A) \tan(45^\circ - A) = 1$$

Solution:

$$\begin{aligned} \text{L.H.S} &= \tan(45^\circ + A) \tan(45^\circ - A) \\ &= \left(\frac{\tan 45^\circ + \tan A}{1 - \tan 45^\circ \tan A} \right) \left(\frac{\tan 45^\circ - \tan A}{1 + \tan 45^\circ \tan A} \right) \\ &= \left(\frac{1 + \tan A}{1 - \tan A} \right) \left(\frac{1 - \tan A}{1 + \tan A} \right) \\ &= 1 = \text{R.H.S} \end{aligned}$$

$$(ii) \quad \tan\left(\frac{\pi}{4} - \theta\right) + \tan\left(\frac{3\pi}{4} + \theta\right) = 0$$

Solution:

$$\text{L.H.S} = \tan\left(\frac{\pi}{4} - \theta\right) + \tan\left(\frac{3\pi}{4} + \theta\right)$$

$$= \left(\frac{\tan \frac{\pi}{4} - \tan \theta}{1 + \tan \frac{\pi}{4} \tan \theta} \right) + \left(\frac{\tan \left(\frac{3\pi}{4} \right) + \tan \theta}{1 - \tan \left(\frac{3\pi}{4} \right) \tan \theta} \right)$$

$$= \left(\frac{1 - \tan \theta}{1 + \tan \theta} \right) + \left(\frac{-1 + \tan \theta}{1 - (-1) \tan \theta} \right)$$

$$= \left(\frac{1 - \tan \theta}{1 + \tan \theta} \right) + \left(\frac{-1 + \tan \theta}{1 + \tan \theta} \right)$$

$$= \frac{1 - \tan \theta - 1 + \tan \theta}{1 + \tan \theta}$$

$$= \frac{0}{1 + \tan \theta}$$

$$= 0 = \text{R.H.S}$$

$$(iii) \quad \sin\left(\theta + \frac{\pi}{6}\right) + \cos\left(\theta + \frac{\pi}{3}\right)$$

$$= \cos \theta$$

Solution:

$$\text{L.H.S} = \sin\left(\theta + \frac{\pi}{6}\right) + \cos\left(\theta + \frac{\pi}{3}\right)$$

$$= \left(\sin \theta \cos \frac{\pi}{6} + \cos \theta \sin \frac{\pi}{6} \right) + \left(\cos \theta \cos \frac{\pi}{3} - \sin \theta \sin \frac{\pi}{3} \right)$$

$$= \sin \theta \frac{\sqrt{3}}{2} + \cos \theta \frac{1}{2} + \cos \theta \frac{1}{2} - \sin \theta \frac{\sqrt{3}}{2}$$

$$= \frac{1}{2} \cos \theta + \frac{1}{2} \cos \theta$$

$$= 2 \times \frac{1}{2} \cos \theta$$

$$= \cos \theta = \text{R.H.S}$$

$$(iv) \quad \frac{\sin \theta - \cos \theta \tan \frac{\theta}{2}}{\cos \theta + \sin \theta \tan \frac{\theta}{2}} = \tan \frac{\theta}{2}$$

Solution:

$$\text{L.H.S} = \frac{\sin \theta - \cos \theta \tan \frac{\theta}{2}}{\cos \theta + \sin \theta \tan \frac{\theta}{2}}$$

$$\begin{aligned}
& \frac{\sin \theta - \cos \theta \times \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}}}{\cos \theta + \sin \theta \times \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}}} \\
&= \frac{\frac{\sin \theta \cos \frac{\theta}{2} - \cos \theta \sin \frac{\theta}{2}}{\cos \frac{\theta}{2}}}{\frac{\cos \theta \cos \frac{\theta}{2} + \sin \theta \sin \frac{\theta}{2}}{\cos \frac{\theta}{2}}} \\
&= \frac{\sin \theta \cos \frac{\theta}{2} - \cos \theta \sin \frac{\theta}{2}}{\cos \theta \cos \frac{\theta}{2} + \sin \theta \sin \frac{\theta}{2}} \\
&= \frac{\sin \left(\theta - \frac{\theta}{2} \right)}{\cos \left(\theta - \frac{\theta}{2} \right)} \\
&= \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} \\
&= \tan \frac{\theta}{2} = \text{R.H.S}
\end{aligned}$$

Alternate solution:

$$\text{L.H.S} = \frac{\sin \theta - \cos \theta \tan \frac{\theta}{2}}{\cos \theta + \sin \theta \tan \frac{\theta}{2}}$$

Dividing up and down by $\cos \theta$

$$\begin{aligned}
& \frac{\sin \theta - \cos \theta \tan \frac{\theta}{2}}{\cos \theta} \\
&= \frac{\sin \theta - \cos \theta \tan \frac{\theta}{2}}{\cos \theta} \\
&= \frac{\sin \theta - \cos \theta \tan \frac{\theta}{2}}{\cos \theta}
\end{aligned}$$

$$\begin{aligned}
& \frac{\frac{\sin \theta}{\cos \theta} - \frac{\cos \theta \tan \frac{\theta}{2}}{\cos \theta}}{\frac{\cos \theta}{\cos \theta} + \frac{\sin \theta \tan \frac{\theta}{2}}{\cos \theta}} \\
&= \frac{\tan \theta - \tan \frac{\theta}{2}}{1 + \tan \theta \tan \frac{\theta}{2}} \\
&= \tan \left(\theta - \frac{\theta}{2} \right) \\
&= \tan \frac{\theta}{2} = \text{R.H.S}
\end{aligned}$$

$$\text{(v)} \quad \frac{1 - \tan \theta \tan \phi}{1 + \tan \theta \tan \phi} = \frac{\cos(\theta + \phi)}{\cos(\theta - \phi)}$$

Solution:

$$\begin{aligned}
\text{L.H.S} &= \frac{1 - \tan \theta \tan \phi}{1 + \tan \theta \tan \phi} \\
&= \frac{\left(1 - \frac{\sin \theta}{\cos \theta} \times \frac{\sin \phi}{\cos \phi} \right)}{\left(1 + \frac{\sin \theta}{\cos \theta} \times \frac{\sin \phi}{\cos \phi} \right)} \\
&= \frac{\frac{\cos \theta \cos \phi - \sin \theta \sin \phi}{\cos \theta \cos \phi}}{\frac{\cos \theta \cos \phi + \sin \theta \sin \phi}{\cos \theta \cos \phi}} \\
&= \frac{\cos \theta \cos \phi - \sin \theta \sin \phi}{\cos \theta \cos \phi + \sin \theta \sin \phi} \\
&= \frac{\cos(\theta + \phi)}{\cos(\theta - \phi)} = \text{R.H.S}
\end{aligned}$$

Q.5 Show that $\cos(\alpha + \beta)\cos(\alpha - \beta) = \cos^2 \alpha - \sin^2 \beta = \cos^2 \beta - \sin^2 \alpha$

Solution:

Consider

$$\begin{aligned}
 & \cos(\alpha + \beta)\cos(\alpha - \beta) \\
 &= (\cos \alpha \cos \beta - \sin \alpha \sin \beta)(\cos \alpha \cos \beta + \sin \alpha \sin \beta) \\
 &= (\cos \alpha \cos \beta)^2 - (\sin \alpha \sin \beta)^2 \\
 &= \cos^2 \alpha \cos^2 \beta - \sin^2 \alpha \sin^2 \beta \\
 &= \cos^2 \alpha (1 - \sin^2 \beta) - (1 - \cos^2 \alpha) \sin^2 \beta \\
 &= \cos^2 \alpha - \cos^2 \alpha \sin^2 \beta - \sin^2 \beta + \cos^2 \alpha \sin^2 \beta \\
 &= \cos^2 \alpha - \sin^2 \beta
 \end{aligned}$$

Further it can be written as

$$\begin{aligned}
 &= (1 - \sin^2 \alpha) - (1 - \cos^2 \beta) \\
 &= 1 - \sin^2 \alpha - 1 + \cos^2 \beta \\
 &= \cos^2 \beta - \sin^2 \alpha
 \end{aligned}$$

Q.6 Show that $\frac{\sin(\alpha + \beta) + \sin(\alpha - \beta)}{\cos(\alpha + \beta) + \cos(\alpha - \beta)} = \tan \alpha$

Solution:

$$\begin{aligned}
 \text{L.H.S} &= \frac{\sin(\alpha + \beta) + \sin(\alpha - \beta)}{\cos(\alpha + \beta) + \cos(\alpha - \beta)} \\
 &= \frac{\sin \alpha \cos \beta + \cancel{\cos \alpha \sin \beta} + \sin \alpha \cos \beta - \cancel{\cos \alpha \sin \beta}}{\cos \alpha \cos \beta - \cancel{\sin \alpha \sin \beta} + \cos \alpha \cos \beta + \cancel{\sin \alpha \sin \beta}} \\
 &= \frac{2 \sin \alpha \cos \beta}{2 \cos \alpha \cos \beta} \\
 &= \frac{\sin \alpha}{\cos \alpha} \\
 &= \tan \alpha = \text{R.H.S}
 \end{aligned}$$

Q.7 Show that:

$$(i) \quad \cot(\alpha + \beta) = \frac{\cot \alpha \cot \beta - 1}{\cot \alpha + \cot \beta}$$

Solution:

$$\begin{aligned} \text{L.H.S} &= \cot(\alpha + \beta) \\ &= \frac{1}{\tan(\alpha + \beta)} \\ &= \frac{1}{\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}} \\ &= \frac{1 - \tan \alpha \tan \beta}{\tan \alpha + \tan \beta} \\ &= \frac{1 - \frac{1}{\cot \alpha \cot \beta}}{\frac{1}{\cot \alpha} + \frac{1}{\cot \beta}} \\ &= \frac{\frac{\cot \alpha \cot \beta - 1}{\cot \alpha \cot \beta}}{\frac{\cot \beta + \cot \alpha}{\cot \alpha \cot \beta}} \\ &= \frac{\cot \alpha \cot \beta - 1}{\cot \alpha + \cot \beta} = \text{R.H.S} \end{aligned}$$

$$(ii) \quad \cot(\alpha - \beta) = \frac{\cot \alpha \cot \beta + 1}{\cot \beta - \cot \alpha}$$

Solution:

$$\begin{aligned} \text{L.H.S} &= \cot(\alpha - \beta) \\ &= \frac{1}{\tan(\alpha - \beta)} \\ &= \frac{1}{\frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}} \end{aligned}$$

$$= \frac{1 + \tan \alpha \tan \beta}{\tan \alpha - \tan \beta}$$

$$= \frac{1 + \frac{1}{\cot \alpha \cot \beta}}{\frac{1}{\cot \alpha} - \frac{1}{\cot \beta}}$$

$$= \frac{\frac{\cot \alpha \cot \beta + 1}{\cot \alpha \cot \beta}}{\frac{\cot \beta - \cot \alpha}{\cot \alpha \cot \beta}}$$

$$= \frac{\cot \alpha \cot \beta + 1}{\cot \beta - \cot \alpha} = \text{R.H.S.}$$

$$(iii) \quad \frac{\tan \alpha + \tan \beta}{\tan \alpha - \tan \beta} = \frac{\sin(\alpha + \beta)}{\sin(\alpha - \beta)}$$

Solution:

$$\begin{aligned} \text{L.H.S} &= \frac{\tan \alpha + \tan \beta}{\tan \alpha - \tan \beta} \\ &= \frac{\frac{\sin \alpha}{\cos \alpha} + \frac{\sin \beta}{\cos \beta}}{\frac{\sin \alpha}{\cos \alpha} - \frac{\sin \beta}{\cos \beta}} \\ &= \frac{\frac{\sin \alpha \cos \beta + \cos \alpha \sin \beta}{\cos \alpha \cos \beta}}{\frac{\sin \alpha \cos \beta - \cos \alpha \sin \beta}{\cos \alpha \cos \beta}} \\ &= \frac{\sin \alpha \cos \beta + \cos \alpha \sin \beta}{\sin \alpha \cos \beta - \cos \alpha \sin \beta} \\ &= \frac{\sin(\alpha + \beta)}{\sin(\alpha - \beta)} = \text{R.H.S} \end{aligned}$$

Q.8 If $\sin \alpha = \frac{4}{5}$ and $\cos \beta = \frac{40}{41}$ where $0 < \alpha < \frac{\pi}{2}$ and $0 < \beta < \frac{\pi}{2}$. Show that

$$\sin(\alpha - \beta) = \frac{133}{205}.$$

Solution:

$$\text{Given that } \sin \alpha = \frac{4}{5}, \cos \beta = \frac{40}{41}$$

$$\text{L.H.S} = \sin(\alpha - \beta)$$

$$= \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$= \sin \alpha \cos \beta - \sqrt{1 - \sin^2 \alpha} \sqrt{1 - \cos^2 \beta} \quad \text{Here } \cos \alpha = +\sqrt{1 - \sin^2 \alpha} \because 0 < \alpha < \frac{\pi}{2}$$

$$= \left(\frac{4}{5}\right)\left(\frac{40}{41}\right) - \sqrt{1 - \frac{16}{25}} \sqrt{1 - \frac{1600}{1681}} \quad \sin \beta = +\sqrt{1 - \cos^2 \beta} \text{ Q } 0 < \beta < \frac{\pi}{2}$$

$$= \frac{160}{205} - \sqrt{\frac{25-16}{25}} \cdot \sqrt{\frac{1681-1600}{1681}}$$

$$= \frac{160}{205} - \sqrt{\frac{9}{25}} \sqrt{\frac{81}{1681}}$$

$$= \frac{160}{205} - \left(\frac{3}{5}\right)\left(\frac{9}{41}\right)$$

$$= \frac{160}{205} - \frac{27}{205}$$

$$= \frac{160-27}{205}$$

$$= \frac{133}{205}$$

Q.9 If $\sin \alpha = \frac{4}{5}$ and $\sin \beta = \frac{12}{13}$, where $\frac{\pi}{2} < \alpha < \pi$ and $\frac{\pi}{2} < \beta < \pi$. Find

(i) $\sin(\alpha + \beta)$

Solution:

$$\text{Given that } \sin \alpha = \frac{4}{5}$$

$$\Rightarrow \cos \alpha = -\sqrt{1 - \sin^2 \alpha}$$

$$\text{Q } \frac{\pi}{2} < \alpha < \pi$$

$$= -\sqrt{1 - \frac{16}{25}}$$

$$= -\sqrt{\frac{25-16}{25}}$$

$$= -\sqrt{\frac{9}{25}}$$

$$\cos \alpha = -\frac{3}{5}$$

$$\text{and } \sin \beta = \frac{4}{5}$$

$$\Rightarrow \cos \beta = -\sqrt{1 - \sin^2 \beta}$$

$$Q \frac{\pi}{2} < \beta < \pi$$

$$= -\sqrt{1 - \frac{144}{169}}$$

$$= -\sqrt{\frac{169 - 144}{169}}$$

$$= -\sqrt{\frac{25}{169}}$$

$$\cos \beta = -\frac{5}{13}$$

$$\text{Using } \sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$= \left(\frac{4}{5}\right)\left(\frac{-5}{13}\right) + \left(\frac{-3}{5}\right)\left(\frac{12}{13}\right)$$

$$= -\frac{20}{65} - \frac{36}{65}$$

$$= \frac{-20 - 36}{65}$$

$$\sin(\alpha + \beta) = \frac{-56}{65}$$

$$(ii) \quad \cos(\alpha + \beta)$$

Solution:

$$\text{Using } \cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$= \left(\frac{-3}{5}\right)\left(\frac{-5}{13}\right) - \left(\frac{4}{5}\right)\left(\frac{12}{13}\right)$$

$$= \frac{15}{65} - \frac{48}{65}$$

$$= \frac{15 - 48}{65}$$

$$\cos(\alpha + \beta) = \frac{-33}{65}$$

(iii) $\tan(\alpha + \beta)$

Solution:

Using $\tan(\alpha + \beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)}$

$$= \frac{\frac{-56}{65}}{\frac{-33}{65}}$$

$$\tan(\alpha + \beta) = \frac{56}{33}$$

Note Here $\sin(\alpha + \beta) = \frac{-56}{65}$ & $\cos(\alpha + \beta) = \frac{-33}{65}$ are both are negative, it means terminal arm of $\alpha + \beta$ lies in III Quadrant

(iv) $\sin(\alpha - \beta)$

Solution:

Using $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$

$$= \left(\frac{4}{5}\right)\left(\frac{-5}{13}\right) - \left(\frac{-3}{5}\right)\left(\frac{12}{13}\right)$$

$$= -\frac{20}{65} + \frac{36}{65}$$

$$= \frac{-20 + 36}{65}$$

$$\sin(\alpha - \beta) = \frac{16}{65}$$

(v) $\cos(\alpha - \beta)$

Solution:

Using $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

$$= \left(\frac{-3}{5}\right)\left(\frac{-5}{13}\right) + \left(\frac{4}{5}\right)\left(\frac{12}{13}\right)$$

$$= \frac{15}{65} + \frac{48}{65}$$

$$\cos(\alpha - \beta) = \frac{63}{65}$$

(vi) $\tan(\alpha - \beta)$

Solution:

Using $\tan(\alpha - \beta) = \frac{\sin(\alpha - \beta)}{\cos(\alpha - \beta)}$

$$= \frac{\frac{16}{65}}{\frac{63}{65}}$$

$$\tan(\alpha - \beta) = \frac{16}{63}$$

Note: Here $\tan(\alpha - \beta) = \frac{16}{63}$ and $\cos(\alpha - \beta) = \frac{63}{65}$ that is both are positive, so terminal arm of $\alpha - \beta$ lies in 1st quadrant.

Q.10 Find $\sin(\alpha + \beta)$ and $\cos(\alpha + \beta)$, given that

(i) $\tan \alpha = \frac{3}{4}$, $\cos \beta = \frac{5}{13}$, and neither the terminal side of the angle of measure α nor that of β is in the I quadrant.

Solution:

It is given that α, β are not in quad. I, it means, α lies in III and β lies in quad. IV.

Here $\tan \alpha = \frac{3}{4}$ and α lies in quad - III

So $\sin \alpha = \frac{-3}{5}$, $\cos \alpha = \frac{-4}{5}$

Also $\cos \beta = \frac{5}{13}$ and β lies in quad - IV

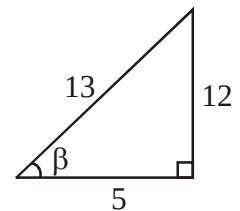
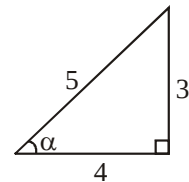
So $\sin \beta = -\frac{12}{13}$

Using $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$

$$= \left(\frac{-3}{5}\right)\left(\frac{5}{13}\right) + \left(\frac{-4}{5}\right)\left(\frac{-12}{13}\right)$$

$$= \frac{-15}{65} + \frac{48}{65}$$

$$\sin(\alpha + \beta) = \frac{33}{65}$$



$$\cos(\alpha + \beta) = ?$$

$$\text{Using } \cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\begin{aligned} &= \left(\frac{-4}{5}\right)\left(\frac{5}{13}\right) - \left(\frac{-3}{5}\right)\left(\frac{-12}{13}\right) \\ &= \frac{-20}{65} - \frac{36}{65} \\ \cos(\alpha + \beta) &= \frac{-56}{65} \end{aligned}$$

(ii) $\tan \alpha = \frac{-15}{8}$ and $\sin \beta = \frac{-7}{25}$ neither the terminal side of the angle of measure α nor that of β is in IV quadrant.

Solution:

It is given that α, β are not in quad. IV, it means, α lies in quad. II and β lies in quad. III.

Here $\tan \alpha = \frac{-15}{8}$ and α lies in quad – II

So $\sin \alpha = \frac{15}{17}$ $\cos \alpha = \frac{-8}{17}$

And for $\sin \beta = \frac{-7}{25}$ and β lies in quad – III

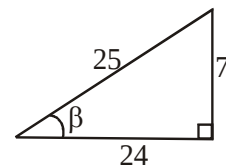
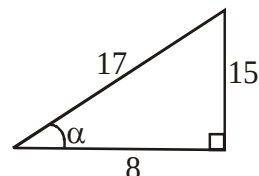
So $\cos \beta = \frac{-24}{25}$

Using $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$

$$\begin{aligned} &= \left(\frac{15}{17}\right)\left(\frac{-24}{25}\right) + \left(\frac{-8}{17}\right)\left(\frac{-7}{25}\right) \\ &= \frac{-360}{425} + \frac{56}{425} \\ &= \frac{-304}{425} \end{aligned}$$

Now

$$\begin{aligned} \cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ &= \left(\frac{-8}{17}\right)\left(\frac{-24}{25}\right) - \left(\frac{15}{17}\right)\left(\frac{-7}{25}\right) \\ &= \frac{192}{425} + \frac{105}{425} \\ &= \frac{297}{425} \end{aligned}$$



Q.11 Prove that $\frac{\cos 8^\circ - \sin 8^\circ}{\cos 8^\circ + \sin 8^\circ} = \tan 37^\circ$

Solution:

$$\text{L.H.S} = \frac{\cos 8^\circ - \sin 8^\circ}{\cos 8^\circ + \sin 8^\circ}$$

Dividing numerator and denominator

by $\cos 8^\circ$, we get

$$\begin{aligned} &= \frac{1 - \frac{\sin 8^\circ}{\cos 8^\circ}}{1 + \frac{\sin 8^\circ}{\cos 8^\circ}} \\ &= \frac{1 - \tan 8^\circ}{1 + \tan 8^\circ} \\ &= \frac{\tan 45^\circ - \tan 8^\circ}{1 + \tan 45^\circ \tan 8^\circ} \\ &= \tan(45^\circ - 8^\circ) \\ &= \tan 37^\circ \\ &= \text{R.H.S} \end{aligned}$$

Alternate Solution:

$$\begin{aligned} \text{R.H.S} &= \tan 37^\circ \\ &= \tan(45^\circ - 8^\circ) \\ &= \frac{\tan 45^\circ - \tan 8^\circ}{1 + \tan 45^\circ \tan 8^\circ} \\ &= \frac{1 - \tan 8^\circ}{1 + \tan 8^\circ} \\ &= \frac{1 - \frac{\sin 8^\circ}{\cos 8^\circ}}{1 + \frac{\sin 8^\circ}{\cos 8^\circ}} \\ &= \frac{\frac{\cos 8^\circ - \sin 8^\circ}{\cos 8^\circ}}{\frac{\cos 8^\circ + \sin 8^\circ}{\cos 8^\circ}} \\ &= \frac{\cos 8^\circ - \sin 8^\circ}{\cos 8^\circ + \sin 8^\circ} = \text{L.H.S} \end{aligned}$$

Q.12 If α, β, γ are the angles of a triangle ABC, show that

$$\cot \frac{\alpha}{2} + \cot \frac{\beta}{2} + \cot \frac{\gamma}{2} = \cot \frac{\alpha}{2} \cot \frac{\beta}{2} \cot \frac{\gamma}{2}$$

Solution:

It is given that α, β, γ are the angles of triangle ABC, so

$$\alpha + \beta + \gamma = 180^\circ$$

$$\Rightarrow \alpha + \beta = 180^\circ - \gamma$$

$$\Rightarrow \frac{\alpha}{2} + \frac{\beta}{2} = 90^\circ - \frac{\gamma}{2}$$

$$\Rightarrow \cot\left(\frac{\alpha}{2} + \frac{\beta}{2}\right) = \cot\left(90^\circ - \frac{\gamma}{2}\right)$$

$$\Rightarrow \frac{\cot \frac{\alpha}{2} \cot \frac{\beta}{2} - 1}{\cot \frac{\alpha}{2} + \cot \frac{\beta}{2}} = \tan \frac{\gamma}{2}$$

$$\Rightarrow \frac{\cot \frac{\alpha}{2} + \cot \frac{\beta}{2}}{\cot \frac{\alpha}{2} \cot \frac{\beta}{2} - 1} = \cot \frac{\gamma}{2}$$

$$\Rightarrow \cot \frac{\alpha}{2} + \cot \frac{\beta}{2} = \cot \frac{\gamma}{2} \left[\cot \frac{\alpha}{2} \cot \frac{\beta}{2} - 1 \right]$$

$$\Rightarrow \cot \frac{\alpha}{2} + \cot \frac{\beta}{2} = \cot \frac{\alpha}{2} \cot \frac{\beta}{2} \cot \frac{\gamma}{2} - \cot \frac{\gamma}{2}$$

$$\Rightarrow \cot \frac{\alpha}{2} + \cot \frac{\beta}{2} + \cot \frac{\gamma}{2} = \cot \frac{\alpha}{2} \cot \frac{\beta}{2} \cot \frac{\gamma}{2}$$

Q.13 If $\alpha + \beta + \gamma = 180^\circ$, show that $\cot \alpha \cot \beta + \cot \beta \cot \gamma + \cot \gamma \cot \alpha = 1$

Solution:

Here $\alpha + \beta + \gamma = 180^\circ$

$$\Rightarrow \alpha + \beta = 180^\circ - \gamma$$

$$\Rightarrow \cot(\alpha + \beta) = \cot(180^\circ - \gamma)$$

$$\Rightarrow \frac{\cot \alpha \cot \beta - 1}{\cot \alpha + \cot \beta} = -\cot \gamma$$

$$\Rightarrow \cot \alpha \cot \beta - 1 = -\cot \gamma [\cot \alpha + \cot \beta]$$

$$\Rightarrow \cot \alpha \cot \beta - 1 = -\cot \gamma \cot \alpha - \cot \gamma \cot \beta$$

$$\Rightarrow \cot \alpha \cot \beta + \cot \beta \cot \gamma + \cot \gamma \cot \alpha = 1$$

Q.14 Express the following in the form of $r \sin(\theta + \phi)$ or $r \sin(\theta - \phi)$, where the terminal sides of the angles of measure θ and ϕ are in the first quadrant.

(i) $12 \sin \theta + 5 \cos \theta$

Solution:

Let $12 = r \cos \phi$ (i)

$5 = r \sin \phi$ (ii)

Squaring (i) & (ii) and then adding, we get,

$$r^2 (\cos^2 \phi + \sin^2 \phi) = 144 + 25$$

$$r^2 = 169$$

$$\Rightarrow r = 13$$

Dividing (ii) by (i) we get,

$$\frac{r \sin \phi}{r \cos \phi} = \frac{5}{12}$$

$$\tan \phi = \frac{5}{12}$$

$$\phi = \tan^{-1} \left(\frac{5}{12} \right)$$

Now given expression can be written as,

$$\begin{aligned} 12 \sin \theta + 5 \cos \theta &= r \cos \phi \sin \theta + r \sin \phi \cos \theta \\ &= r (\sin \theta \cos \phi + \cos \theta \sin \phi) \\ &= r \sin (\theta + \phi) \end{aligned}$$

$$\text{Where } r = 13 \text{ and } \phi = \tan^{-1} \frac{5}{12}$$

$$(ii) \quad 3 \sin \theta - 4 \cos \theta$$

Solution:

$$\text{Let } 3 = r \cos \phi \tag{i}$$

$$4 = r \sin \phi \tag{ii}$$

Squaring (i) and (ii) and then adding, we get,

$$r^2 (\cos^2 \phi + \sin^2 \phi) = 9 + 16$$

$$r^2 = 25$$

$$r = 5$$

Dividing (ii) by (i) we get,

$$\frac{r \sin \phi}{r \cos \phi} = \frac{4}{3}$$

$$\tan \phi = \frac{4}{3}$$

$$\phi = \tan^{-1} \frac{4}{3}$$

Given expression can be written as,

$$\begin{aligned} 3 \sin \theta - 4 \cos \theta &= r \cos \phi \sin \theta - r \sin \phi \cos \theta \\ &= r (\sin \theta \cos \phi - \cos \theta \sin \phi) \\ &= r \sin (\theta - \phi) \end{aligned}$$

$$\text{Where } r = 5 \text{ and } \phi = \tan^{-1} \frac{4}{3}$$

(iii) $\sin\theta - \cos\theta$ **Solution:**

$$\text{Let } r \cos \phi = 1 \quad (i)$$

$$r \sin \phi = 1 \quad (ii)$$

Squaring (i) and (ii) and then adding, we get,

$$r^2 (\cos^2 \phi + \sin^2 \phi) = 1^2 + 1^2$$

$$r^2 = 2$$

$$\Rightarrow r = \sqrt{2}$$

Dividing (ii) by (i) we get .

$$\frac{r \sin \phi}{r \cos \phi} = \frac{1}{1}$$

$$\tan \phi = 1$$

$$\Rightarrow \phi = \tan^{-1} 1$$

Given expression can be written as,

$$\begin{aligned} \sin \theta - \cos \theta &= r \cos \phi \sin \theta - r \sin \phi \cos \theta \\ &= r (\sin \theta \cos \phi - \cos \theta \sin \phi) \\ &= r \sin (\theta - \phi) \end{aligned}$$

$$\text{Where } r = \sqrt{2} \text{ and } \phi = \tan^{-1} 1$$

(iv) $5\sin\theta - 4\cos\theta$ **Solution:**

$$\text{Let } r \cos \phi = 5 \quad (i)$$

$$r \sin \phi = 4 \quad (ii)$$

Squaring (i) & (ii) and then adding, we get,

$$r^2 (\cos^2 \phi + \sin^2 \phi) = 25 + 16$$

$$r^2 = 41$$

$$\Rightarrow r = \sqrt{41}$$

Dividing (ii) by (i) we get,

$$\frac{r \sin \phi}{r \cos \phi} = \frac{4}{5}$$

$$\tan \phi = \frac{4}{5}$$

$$\Rightarrow \phi = \tan^{-1} \frac{4}{5}$$

Given expression can be written as,

$$\begin{aligned} 5\sin\theta - 4\cos\theta &= r\cos\phi\sin\theta - r\sin\phi\cos\theta \\ &= r(\sin\theta\cos\phi - \cos\theta\sin\phi) \\ &= r\sin(\theta - \phi) \end{aligned}$$

$$\text{Where } r = \sqrt{41} \text{ and } \phi = \tan^{-1} \frac{4}{5}$$

(v) $\sin\theta + \cos\theta$

Solution:

$$\text{Let } r\cos\phi = 1 \quad (i)$$

$$r\sin\phi = 1 \quad (ii)$$

Squaring (i) and (ii) and then adding, we get,

$$r^2(\cos^2\phi + \sin^2\phi) = 1^2 + 1^2$$

$$r^2 = 2$$

$$\Rightarrow r = \sqrt{2}$$

Dividing (ii) by (i) we get,

$$\frac{r\sin\phi}{r\cos\phi} = \frac{1}{1}$$

$$\tan\phi = 1$$

$$\Rightarrow \phi = \tan^{-1}(1)$$

Given expression can be written as,

$$\begin{aligned} \sin\theta + \cos\theta &= r\cos\phi\sin\theta + r\sin\phi\cos\theta \\ &= r(\sin\theta\cos\phi + \cos\theta\sin\phi) \\ &= r\sin(\theta + \phi) \end{aligned}$$

$$\text{Where } r = \sqrt{2} \text{ and } \phi = \tan^{-1} 1$$

(vi) $3\sin\theta - 5\cos\theta$

Solution:

$$\text{Let } r\cos\phi = 3 \quad (i)$$

$$r\sin\phi = 5 \quad (ii)$$

Squaring (i) and (ii) and then adding we get,

$$r^2(\cos^2\phi + \sin^2\phi) = 3^2 + 5^2$$

$$r^2 = 9 + 25$$

$$\Rightarrow r = \sqrt{34}$$

Dividing (ii) by (i) we get,

$$\frac{r \sin \phi}{r \cos \phi} = \frac{5}{3}$$

$$\tan \phi = \frac{5}{3}$$

$$\Rightarrow \phi = \tan^{-1} \frac{5}{3}$$

Given expression can be written as,

$$\begin{aligned} 3 \sin \theta - 5 \cos \theta &= r \cos \phi \sin \theta - r \sin \phi \cos \theta \\ &= r (\sin \theta \cos \phi - \cos \theta \sin \phi) \\ &= r \sin (\theta - \phi) \end{aligned}$$

$$\text{Where } r = \sqrt{34} \text{ and } \phi = \tan^{-1} \frac{5}{3}$$

Double Angle Identities:

(i) $\sin 2\alpha = 2 \sin \alpha \cos \alpha$

(ii) $\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$
 $= 2 \cos^2 \alpha - 1$
 $= 1 - 2 \sin^2 \alpha$

(iii) $\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$

Proof:

(i)

We know that

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

Replace ' β ' by ' α ' we get

$$\sin(\alpha + \alpha) = \sin \alpha \cos \alpha + \cos \alpha \sin \alpha$$

$$\text{Hence } \boxed{\sin 2\alpha = 2 \sin \alpha \cos \alpha}$$

(ii)

We know that

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

Replace β by α

$$\cos(\alpha + \alpha) = \cos \alpha \cos \alpha - \sin \alpha \sin \alpha$$

$$\text{Hence } \boxed{\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha}$$

Now

$$\begin{aligned} \cos 2\alpha &= \cos^2 \alpha - \sin^2 \alpha \\ &= \cos^2 \alpha - (1 - \cos^2 \alpha) \\ &= \cos^2 \alpha - 1 + \cos^2 \alpha \end{aligned}$$

$$\text{Hence } \boxed{\cos 2\alpha = 2 \cos^2 \alpha - 1}$$

$$\begin{aligned} \text{Again } \cos 2\alpha &= \cos^2 \alpha - \sin^2 \alpha \\ &= 1 - \sin^2 \alpha - \sin^2 \alpha \end{aligned}$$

$$\text{Hence } \boxed{\cos 2\alpha = 1 - 2 \sin^2 \alpha}$$

(iii)

We know that

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

Replace ' β ' by ' α ' we get.

$$\tan(\alpha + \alpha) = \frac{\tan \alpha + \tan \alpha}{1 - \tan \alpha \tan \alpha}$$

$$\text{Hence } \boxed{\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}}$$

Half Angle Identities:

$$(i) \quad \cos \frac{\alpha}{2} = \pm \sqrt{\frac{1 + \cos \alpha}{2}}$$

$$(ii) \quad \sin \frac{\alpha}{2} = \pm \sqrt{\frac{1 - \cos \alpha}{2}}$$

$$(iii) \quad \tan \frac{\alpha}{2} = \pm \sqrt{\frac{1 - \cos \alpha}{1 + \cos \alpha}}$$

Proof:**(i)**

We know that

$$\cos 2\theta = 2\cos^2 \theta - 1$$

Replace " θ " by " $\frac{\alpha}{2}$ " we get.

$$\cos 2\left(\frac{\alpha}{2}\right) = 2\cos^2\left(\frac{\alpha}{2}\right) - 1$$

$$\cos \alpha = 2\cos^2 \frac{\alpha}{2} - 1$$

$$1 + \cos \alpha = 2\cos^2 \frac{\alpha}{2}$$

$$\frac{1 + \cos \alpha}{2} = \cos^2 \frac{\alpha}{2}$$

$$\pm \sqrt{\frac{1 + \cos \alpha}{2}} = \cos \frac{\alpha}{2}$$

$$\text{Hence } \boxed{\cos \frac{\alpha}{2} = \pm \sqrt{\frac{1 + \cos \alpha}{2}}}$$

(ii)

We know that

$$\cos 2\theta = 1 - 2\sin^2 \theta$$

Replace " θ " by " $\frac{\alpha}{2}$ " we get.

$$\cos 2\left(\frac{\alpha}{2}\right) = 1 - 2\sin^2 \frac{\alpha}{2}$$

$$\cos \alpha = 1 - 2\sin^2 \frac{\alpha}{2}$$

$$2\sin^2 \frac{\alpha}{2} = 1 - \cos \alpha$$

$$\sin^2 \frac{\alpha}{2} = \frac{1 - \cos \alpha}{2}$$

$$\text{Hence } \boxed{\sin \frac{\alpha}{2} = \pm \sqrt{\frac{1 - \cos \alpha}{2}}}$$

(iii)

We know that

$$\tan \frac{\alpha}{2} = \frac{\sin \frac{\alpha}{2}}{\cos \frac{\alpha}{2}}$$

$$= \pm \frac{\sqrt{\frac{1 - \cos \alpha}{2}}}{\sqrt{\frac{1 + \cos \alpha}{2}}}$$

$$\tan \frac{\alpha}{2} = \pm \sqrt{\frac{1 - \cos \alpha}{1 + \cos \alpha}}$$

$$\text{Hence } \boxed{\tan \frac{\alpha}{2} = \pm \sqrt{\frac{1 - \cos \alpha}{1 + \cos \alpha}}}$$

Triple angle identities:

$$(i) \quad \sin 3\alpha = 3\sin\alpha - 4\sin^3\alpha$$

$$(ii) \quad \cos 3\alpha = 4\cos^3\alpha - 3\cos\alpha$$

$$(iii) \quad \tan 3\alpha = \frac{3\tan\alpha - \tan^3\alpha}{1 - 3\tan^2\alpha}$$

Proof:**(i)**

$$\begin{aligned} \text{L.H.S } \sin 3\alpha &= \sin(2\alpha + \alpha) \\ &= \sin 2\alpha \cos \alpha + \cos 2\alpha \sin \alpha \\ &= (2\sin \alpha \cos \alpha) \cos \alpha + (1 - 2\sin^2 \alpha) \sin \alpha \\ &= 2\sin \alpha \cos^2 \alpha + \sin \alpha - 2\sin^3 \alpha \\ &= 2\sin \alpha (1 - \sin^2 \alpha) + \sin \alpha - 2\sin^3 \alpha \\ &= 2\sin \alpha - 2\sin^3 \alpha + \sin \alpha - 2\sin^3 \alpha \end{aligned}$$

$$\text{Hence } \boxed{\sin 3\alpha = 3\sin \alpha - 4\sin^3 \alpha}$$

(ii)

$$\begin{aligned} \text{L.H.S } \cos 3\alpha &= \cos(2\alpha + \alpha) \\ &= \cos 2\alpha \cos \alpha - \sin 2\alpha \sin \alpha \\ &= (2\cos^2 \alpha - 1) \cos \alpha - (2\sin \alpha \cos \alpha) \sin \alpha \\ &= 2\cos^3 \alpha - \cos \alpha - 2\sin^2 \alpha \cos \alpha \\ &= 2\cos^3 \alpha - \cos \alpha - 2(1 - \cos^2 \alpha) \cos \alpha \\ &= 2\cos^3 \alpha - \cos \alpha - 2\cos \alpha + 2\cos^3 \alpha \end{aligned}$$

$$\text{Hence } \boxed{\cos 3\alpha = 4\cos^3 \alpha - 3\cos \alpha}$$

(iii)

$$\begin{aligned} \text{L.H.S } \tan 3\alpha &= \tan(2\alpha + \alpha) \\ &= \frac{\tan 2\alpha + \tan \alpha}{1 - \tan 2\alpha \tan \alpha} \\ &= \frac{\frac{2\tan \alpha}{1 - \tan^2 \alpha} + \tan \alpha}{1 - \frac{2\tan \alpha}{1 - \tan^2 \alpha} \cdot \tan \alpha} \\ &= \frac{\frac{2\tan \alpha + \tan \alpha(1 - \tan^2 \alpha)}{1 - \tan^2 \alpha}}{\frac{(1 - \tan^2 \alpha) - 2\tan \alpha \cdot \tan \alpha}{1 - \tan^2 \alpha}} \\ &= \frac{2\tan \alpha + \tan \alpha - \tan^3 \alpha}{1 - \tan^2 \alpha - 2\tan^2 \alpha} \end{aligned}$$

$$\text{Hence } \boxed{\tan 3\alpha = \frac{3\tan \alpha - \tan^3 \alpha}{1 - 3\tan^2 \alpha}}$$

Exercise 10.3

Q.1 Find the values of $\sin 2\alpha$, $\cos 2\alpha$ and $\tan 2\alpha$, when:

(i) $\sin \alpha = \frac{12}{13}$

Where $0 < \alpha < \frac{\pi}{2}$

Solution:

Given that: $\sin \alpha = \frac{12}{13}$

$$\Rightarrow \cos \alpha = \sqrt{1 - \sin^2 \alpha}$$

$$= \sqrt{1 - \left(\frac{12}{13}\right)^2}$$

$$= \sqrt{1 - \frac{144}{169}}$$

$$= \sqrt{\frac{169 - 144}{169}}$$

$$= \sqrt{\frac{25}{169}}$$

$$\cos \alpha = \frac{5}{13}$$

Using $\sin 2\alpha = 2 \sin \alpha \cos \alpha$

$$= 2 \left(\frac{12}{13}\right) \left(\frac{5}{13}\right)$$

$$\sin 2\alpha = \frac{120}{169}$$

Now $\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$

$$\cos 2\alpha = \left(\frac{5}{13}\right)^2 - \left(\frac{12}{13}\right)^2$$

$$= \frac{25}{169} - \frac{144}{169}$$

$$= \frac{25 - 144}{169}$$

$$\cos 2\alpha = \frac{-119}{169}$$

$$\text{Now } \tan 2\alpha = \frac{\sin 2\alpha}{\cos 2\alpha} = \frac{\frac{120}{169}}{\frac{-119}{169}}$$

$$\tan 2\alpha = \frac{-120}{119}$$

(ii) $\cos \alpha = \frac{3}{5}$

Solution:

Given that: $\cos \alpha = \frac{3}{5}$

$$\sin \alpha = \sqrt{1 - \cos^2 \alpha}$$

$$= \sqrt{1 - \left(\frac{3}{5}\right)^2}$$

$$= \sqrt{1 - \frac{9}{25}}$$

$$= \sqrt{\frac{25 - 9}{25}}$$

$$= \sqrt{\frac{16}{25}}$$

$$\sin \alpha = \frac{4}{5}$$

Using $\sin 2\alpha = 2 \sin \alpha \cos \alpha$

$$= 2 \left(\frac{4}{5}\right) \left(\frac{3}{5}\right)$$

$$\sin 2\alpha = \frac{24}{25}$$

Now $\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$

$$= \left(\frac{3}{5}\right)^2 - \left(\frac{4}{5}\right)^2$$

$$= \frac{9}{25} - \frac{16}{25}$$

$$= \frac{9 - 16}{25}$$

$$\cos 2\alpha = \frac{-7}{25}$$

$$\text{Now } \tan 2\alpha = \frac{\sin 2\alpha}{\cos 2\alpha} = \frac{\frac{24}{25}}{\frac{-7}{25}}$$

$$\tan 2\alpha = \frac{-24}{7}$$

Prove the following Identities:

Q.2 $\cot \alpha - \tan \alpha = 2 \cot 2\alpha$

Solution:

$$\begin{aligned} \text{L.H.S.} &= \cot \alpha - \tan \alpha \\ &= \frac{\cos \alpha}{\sin \alpha} - \frac{\sin \alpha}{\cos \alpha} \\ &= \frac{\cos^2 \alpha - \sin^2 \alpha}{\sin \alpha \cos \alpha} \\ &= \frac{\cos 2\alpha}{\sin \alpha \cos \alpha} \\ &= \frac{2 \cos 2\alpha}{2 \sin \alpha \cos \alpha} \\ &= \frac{2 \cos 2\alpha}{\sin 2\alpha} \\ &= 2 \cot 2\alpha = \text{R.H.S.} \end{aligned}$$

Q.3 $\frac{\sin 2\alpha}{1 + \cos 2\alpha} = \tan \alpha$

Solution:

$$\begin{aligned} \text{L.H.S.} &= \frac{\sin 2\alpha}{1 + \cos 2\alpha} \\ &= \frac{2 \sin \alpha \cos \alpha}{2 \cos^2 \alpha} \\ &= \frac{\sin \alpha}{\cos \alpha} \\ &= \tan \alpha = \text{R.H.S.} \end{aligned}$$

Q.4 $\frac{1 - \cos \alpha}{\sin \alpha} = \tan \frac{\alpha}{2}$

Solution:

$$\begin{aligned} \text{L.H.S.} &= \frac{1 - \cos \alpha}{\sin \alpha} \\ &= \frac{2 \sin^2 \frac{\alpha}{2}}{2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2}} \end{aligned}$$

$$\begin{aligned} &= \frac{\sin \frac{\alpha}{2}}{\cos \frac{\alpha}{2}} \\ &= \tan \frac{\alpha}{2} = \text{R.H.S.} \end{aligned}$$

Q.5 $\frac{\cos \alpha - \sin \alpha}{\cos \alpha + \sin \alpha} = \sec 2\alpha - \tan 2\alpha$

Solution:

$$\begin{aligned} \text{R.H.S.} &= \sec 2\alpha - \tan 2\alpha \\ &= \frac{1}{\cos 2\alpha} - \frac{\sin 2\alpha}{\cos 2\alpha} \\ &= \frac{1 - \sin 2\alpha}{\cos 2\alpha} \\ &= \frac{\sin^2 \alpha + \cos^2 \alpha - 2 \sin \alpha \cos \alpha}{\cos^2 \alpha - \sin^2 \alpha} \\ &= \frac{(\cos \alpha - \sin \alpha)^2}{(\cos \alpha - \sin \alpha)(\cos \alpha + \sin \alpha)} \\ &= \frac{\cos \alpha - \sin \alpha}{\cos \alpha + \sin \alpha} = \text{L.H.S.} \end{aligned}$$

Alternate Solution:

$$\begin{aligned} \text{L.H.S.} &= \frac{\cos \alpha - \sin \alpha}{\cos \alpha + \sin \alpha} \\ &= \frac{\cos \alpha - \sin \alpha}{\cos \alpha + \sin \alpha} \times \frac{\cos \alpha - \sin \alpha}{\cos \alpha - \sin \alpha} \\ &= \frac{(\cos \alpha - \sin \alpha)^2}{\cos^2 \alpha - \sin^2 \alpha} \\ &= \frac{\cos^2 \alpha + \sin^2 \alpha - 2 \sin \alpha \cos \alpha}{\cos 2\alpha} \\ &= \frac{1 - \sin 2\alpha}{\cos 2\alpha} \\ &= \frac{1}{\cos 2\alpha} - \frac{\sin 2\alpha}{\cos 2\alpha} = \sec 2\alpha - \tan 2\alpha \\ &= \text{R.H.S.} \end{aligned}$$

$$\text{Q.6} \quad \sqrt{\frac{1+\sin\alpha}{1-\sin\alpha}} = \frac{\sin\frac{\alpha}{2} + \cos\frac{\alpha}{2}}{\sin\frac{\alpha}{2} - \cos\frac{\alpha}{2}}$$

Solution:

$$\begin{aligned} \text{L.H.S.} &= \sqrt{\frac{1+\sin\alpha}{1-\sin\alpha}} \\ &= \sqrt{\frac{\sin^2\frac{\alpha}{2} + \cos^2\frac{\alpha}{2} + 2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}}{\sin^2\frac{\alpha}{2} + \cos^2\frac{\alpha}{2} - 2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}}} \\ &= \sqrt{\frac{\left(\sin\frac{\alpha}{2} + \cos\frac{\alpha}{2}\right)^2}{\left(\sin\frac{\alpha}{2} - \cos\frac{\alpha}{2}\right)^2}} \\ &= \frac{\sin\frac{\alpha}{2} + \cos\frac{\alpha}{2}}{\sin\frac{\alpha}{2} - \cos\frac{\alpha}{2}} = \text{R.H.S.} \end{aligned}$$

$$\text{Q.7} \quad \frac{\operatorname{cosec}\theta + 2\operatorname{cosec}2\theta}{\sec\theta} = \cot\frac{\theta}{2}$$

Solution:

$$\begin{aligned} \text{L.H.S.} &= \frac{\operatorname{cosec}\theta + 2\operatorname{cosec}2\theta}{\sec\theta} \\ &= \frac{\operatorname{cosec}\theta}{\sec\theta} + \frac{2\operatorname{cosec}2\theta}{\sec\theta} \\ &= \frac{\cos\theta}{\sin\theta} + \frac{2\cos\theta}{\sin 2\theta} \\ &= \frac{\cos\theta}{\sin\theta} + \frac{2\cos\theta}{2\sin\theta\cos\theta} \\ &= \frac{\cos\theta}{\sin\theta} + \frac{1}{\sin\theta} \\ &= \frac{1+\cos\theta}{\sin\theta} \\ &= \frac{2\cos^2\frac{\theta}{2}}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}} \end{aligned}$$

$$\begin{aligned} &= \frac{\cos\frac{\theta}{2}}{\sin\frac{\theta}{2}} \\ &= \cot\frac{\theta}{2} = \text{R.H.S.} \end{aligned}$$

$$\text{Q.8} \quad 1 + \tan\alpha \tan 2\alpha = \sec 2\alpha$$

Solution:

$$\begin{aligned} \text{L.H.S.} &= 1 + \tan\alpha \tan 2\alpha \\ &= 1 + \frac{\sin\alpha}{\cos\alpha} \cdot \frac{\sin 2\alpha}{\cos 2\alpha} \\ &= \frac{\cos 2\alpha \cos\alpha + \sin 2\alpha \cdot \sin\alpha}{\cos\alpha \cos 2\alpha} \\ &= \frac{\cos(2\alpha - \alpha)}{\cos\alpha \cos 2\alpha} \\ &= \frac{\cos\alpha}{\cos\alpha \cos 2\alpha} \\ &= \frac{1}{\cos 2\alpha} \\ &= \sec 2\alpha = \text{R.H.S.} \end{aligned}$$

$$\text{Q.9} \quad \frac{2\sin\theta \sin 2\theta}{\cos\theta + \cos 3\theta} = \tan 2\theta \tan \theta$$

Solution:

$$\begin{aligned} \text{L.H.S.} &= \frac{2\sin\theta \sin 2\theta}{\cos\theta + \cos 3\theta} \\ &= \frac{2\sin\theta \sin 2\theta}{\cos\theta + (4\cos^3\theta - 3\cos\theta)} \\ &= \frac{2\sin\theta \sin 2\theta}{\cos\theta [1 + 4\cos^2\theta - 3]} \\ &= \frac{2\sin\theta \sin 2\theta}{\cos\theta (4\cos^2\theta - 2)} \\ &= \frac{2\sin\theta \sin 2\theta}{2\cos\theta (2\cos^2\theta - 1)} \end{aligned}$$

$$\begin{aligned}
 &= \frac{2 \sin \theta \sin 2\theta}{2 \cos \theta \cdot \cos 2\theta} \\
 &= \frac{\sin \theta}{\cos \theta} \cdot \frac{\sin 2\theta}{\cos 2\theta} \\
 &= \tan \theta \cdot \tan 2\theta = \text{R.H.S.}
 \end{aligned}$$

Q.10 $\frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta} = 2$

Solution:

$$\begin{aligned}
 \text{L.H.S} &= \frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta} \\
 &= \frac{\sin 3\theta \cos \theta - \cos 3\theta \sin \theta}{\sin \theta \cos \theta} \\
 &= \frac{\sin(3\theta - \theta)}{\sin \theta \cos \theta} \\
 &= \frac{\sin 2\theta}{\sin \theta \cos \theta} \\
 &= \frac{2 \sin \theta \cos \theta}{\sin \theta \cos \theta} \\
 &= 2 = \text{R.H.S.}
 \end{aligned}$$

Q.11 $\frac{\cos 3\theta}{\cos \theta} + \frac{\sin 3\theta}{\sin \theta} = 4 \cos 2\theta$

Solution:

$$\begin{aligned}
 \text{L.H.S} &= \frac{\cos 3\theta}{\cos \theta} + \frac{\sin 3\theta}{\sin \theta} \\
 &= \frac{\cos 3\theta \sin \theta + \sin 3\theta \cos \theta}{\cos \theta \sin \theta} \\
 &= \frac{\sin 3\theta \cos \theta + \cos 3\theta \sin \theta}{\sin \theta \cos \theta} \\
 &= \frac{\sin(3\theta + \theta)}{\cos \theta \sin \theta} \\
 &= \frac{\sin 4\theta}{\cos \theta \sin \theta} \\
 &= \frac{2 \sin 2\theta \cos 2\theta}{\cos \theta \sin \theta}
 \end{aligned}$$

$\therefore$ multiply and divide by 2

$$\begin{aligned}
 &= \frac{2 \times 2 \sin 2\theta \cos 2\theta}{2 \cos \theta \sin \theta} \\
 &= \frac{4 \sin 2\theta \cos 2\theta}{\sin 2\theta} \\
 &= 4 \cos 2\theta = \text{R.H.S}
 \end{aligned}$$

Q.12 $\frac{\tan \frac{\theta}{2} + \cot \frac{\theta}{2}}{\cot \frac{\theta}{2} - \tan \frac{\theta}{2}} = \sec \theta$

Solution:

$$\begin{aligned}
 \text{L.H.S} &= \frac{\tan \frac{\theta}{2} + \cot \frac{\theta}{2}}{\cot \frac{\theta}{2} - \tan \frac{\theta}{2}} \\
 &= \frac{\frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} + \frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}}}{\frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} - \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}}} \\
 &= \frac{\frac{\sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2}}{\sin \frac{\theta}{2} \cos \frac{\theta}{2}}}{\frac{\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}}{\sin \frac{\theta}{2} \cos \frac{\theta}{2}}} \\
 &= \frac{\sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2}}{\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}} \\
 &= \frac{1}{\cos \theta} = \sec \theta = \text{R.H.S.}
 \end{aligned}$$

Q.13 $\frac{\sin 3\theta}{\cos \theta} + \frac{\cos 3\theta}{\sin \theta} = 2 \cot 2\theta$

Solution:

$$\begin{aligned} \text{L.H.S.} &= \frac{\sin 3\theta}{\cos \theta} + \frac{\cos 3\theta}{\sin \theta} \\ &= \frac{\sin 3\theta \sin \theta + \cos 3\theta \cos \theta}{\cos \theta \sin \theta} \\ &= \frac{\cos(3\theta - \theta)}{\cos \theta \sin \theta} \\ &= \frac{\cos 2\theta}{\cos \theta \sin \theta} \\ &= \frac{2 \cos 2\theta}{2 \sin \theta \cos \theta} \\ &= \frac{2 \cos 2\theta}{\sin 2\theta} \\ &= 2 \cot 2\theta = \text{R.H.S.} \end{aligned}$$

Q.14 Reduce $\sin^4 \theta$ to an expression involving only function of multiples of θ raised to the first power.

Solution:

Here

$$\begin{aligned} \sin^4 \theta &= (\sin^2 \theta)^2 \\ &= \left(\frac{1 - \cos 2\theta}{2} \right)^2 \\ &= \frac{1}{4} (1 - \cos 2\theta)^2 \\ &= \frac{1}{4} (1 + \cos^2 2\theta - 2 \cos 2\theta) \\ &= \frac{1}{4} \left(1 + \frac{1 + \cos 4\theta}{2} - 2 \cos 2\theta \right) \\ &= \frac{1}{4} \left(1 + \frac{1}{2} + \frac{\cos 4\theta}{2} - 2 \cos 2\theta \right) \\ &= \frac{1}{4} \left(\frac{3}{2} + \frac{\cos 4\theta}{2} - 2 \cos 2\theta \right) \\ &= \frac{3}{8} + \frac{\cos 4\theta}{8} - \frac{2 \cos 2\theta}{4} \\ &= \frac{3 + \cos 4\theta - 4 \cos 2\theta}{8} \\ &= \frac{3 - 4 \cos 2\theta + \cos 4\theta}{8} \end{aligned}$$

Q.15 Find the value of $\sin \theta$ and $\cos \theta$ without using table or calculator, when θ is
(i) 18° (ii) 36° (iii) 54° (iv) 72°

Hence prove that: $\cos 36^\circ \cos 72^\circ \cos 108^\circ \cos 144^\circ = \frac{1}{16}$

Solution:

(i)

$$\begin{aligned} \text{Let } \theta &= 18^\circ \\ \Rightarrow 5\theta &= 90^\circ \\ \Rightarrow 3\theta + 2\theta &= 90^\circ \\ \Rightarrow 3\theta &= 90^\circ - 2\theta \\ \Rightarrow \sin 3\theta &= \sin(90^\circ - 2\theta) \\ 3\sin \theta - 4\sin^3 \theta &= \cos 2\theta \end{aligned}$$

$$3\sin\theta - 4\sin^3\theta = 1 - 2\sin^2\theta$$

$$4\sin^3\theta - 2\sin^2\theta - 3\sin\theta + 1 = 0$$

Using synthetic division here

1	4	-2	-3	1
		4	2	-1
4	2	-1	0	

So one of the root of equation is

$$\sin\theta = 1 \Rightarrow \theta = 90^\circ \text{ (neglect it because our } \theta = 18^\circ \text{)}$$

Depressed equation is

$$4\sin^2\theta + 2\sin\theta - 1 = 0$$

Here $a = 4, b = 2, c = -1$

$$\sin\theta = \frac{-2 \pm \sqrt{4 - 4(4)(-1)}}{2(4)}$$

$$= \frac{-2 \pm \sqrt{4 + 16}}{8}$$

$$= \frac{-2 \pm \sqrt{20}}{8}$$

$$= \frac{-2 \pm 2\sqrt{5}}{8}$$

$$= \frac{-1 \pm \sqrt{5}}{4}$$

$$\sin\theta = \frac{\sqrt{5}-1}{4} > 0, \quad \sin\theta = \frac{-\sqrt{5}-1}{4} < 0$$

As here $\theta = 18^\circ$ lies in 1st quadrant, so $\sin\theta$ should be positive.

So, $\boxed{\sin 18^\circ = \frac{\sqrt{5}-1}{4}}$

Now $\cos 72^\circ = \cos(90^\circ - 18^\circ)$

$$= \sin 18^\circ$$

$$\boxed{\cos 72^\circ = \sin 18^\circ = \frac{\sqrt{5}-1}{4}}$$

Next let $\theta = 36^\circ$

$$\Rightarrow 5\theta = 36^\circ \times 5$$

$$5\theta = 180^\circ$$

$$3\theta + 2\theta = 180^\circ$$

$$3\theta = 180^\circ - 2\theta$$

$$\sin 3\theta = \sin(180^\circ - 2\theta)$$

$$3\sin \theta - 4\sin^3 \theta = \sin 2\theta$$

$$3\sin \theta - 4\sin^3 \theta - 2\sin \theta \cos \theta = 0$$

$$\sin \theta (3 - 4\sin^2 \theta - 2\cos \theta) = 0$$

$$\sin \theta \neq 0 \text{ because here } \theta = 36^\circ \text{ so } \sin 36^\circ \neq 0$$

$$\text{and } 3 - 4\sin^2 \theta - 2\cos \theta = 0$$

$$4\sin^2 \theta + 2\cos \theta - 3 = 0$$

$$4(1 - \cos^2 \theta) + 2\cos \theta - 3 = 0$$

$$4 - 4\cos^2 \theta + 2\cos \theta - 3 = 0$$

$$-4\cos^2 \theta + 2\cos \theta + 1 = 0$$

$$4\cos^2 \theta - 2\cos \theta - 1 = 0$$

$$\text{Here } a = 4, b = -2, c = -1$$

$$\text{Using } \cos \theta = \frac{2 \pm \sqrt{4 - 4(4)(-1)}}{2(4)}$$

$$= \frac{2 \pm \sqrt{4 + 16}}{8}$$

$$= \frac{2 \pm \sqrt{20}}{8}$$

$$= \frac{2 \pm 2\sqrt{5}}{8}$$

$$= \frac{1 \pm \sqrt{5}}{4}$$

$$\text{Here } \cos \theta = \frac{1 - \sqrt{5}}{4} < 0 \text{ and } \cos \theta = \frac{1 + \sqrt{5}}{4} > 0$$

So neglect $\cos \theta = \frac{1 - \sqrt{5}}{4} < 0$ because here $\theta = 36^\circ$ lies in 1st quadrant so $\cos 36^\circ$ should be positive.

$$\text{So } \boxed{\cos 36^\circ = \frac{1+\sqrt{5}}{4}}$$

$$\begin{aligned}\text{Also } \cos 36^\circ &= \cos(90^\circ - 54^\circ) \\ &= \sin 54^\circ\end{aligned}$$

$$\therefore \boxed{\cos 36^\circ = \sin 54^\circ = \frac{1+\sqrt{5}}{4}}$$

$$\begin{aligned}\text{Now } \cos^2 18^\circ &= 1 - \sin^2 18^\circ \\ &= 1 - \left(\frac{\sqrt{5}-1}{4}\right)^2 \\ &= 1 - \left(\frac{5+1-2\sqrt{5}}{16}\right) \\ &= \frac{16 - (6 - 2\sqrt{5})}{16}\end{aligned}$$

$$\cos^2 18^\circ = \frac{10 + 2\sqrt{5}}{16}$$

$$\Rightarrow \cos 18^\circ = \pm \sqrt{\frac{10 + 2\sqrt{5}}{16}}$$

$\therefore \theta = 18^\circ$ lies in quad. I so $\cos 18^\circ > 0$

$$\boxed{\cos 18^\circ = \frac{\sqrt{10 + 2\sqrt{5}}}{4}}$$

$$\begin{aligned}\text{Also } \cos 18^\circ &= \cos(90^\circ - 72^\circ) \\ &= \sin 72^\circ\end{aligned}$$

$$\therefore \boxed{\cos 18^\circ = \sin 72^\circ = \frac{\sqrt{10 + 2\sqrt{5}}}{4}}$$

$$\begin{aligned}\text{Now } \cos^2 54^\circ &= 1 - \sin^2 54^\circ \\ &= 1 - \left(\frac{1+\sqrt{5}}{4}\right)^2 \\ &= 1 - \left(\frac{1+5+2\sqrt{5}}{16}\right)\end{aligned}$$

$$\begin{aligned}
 &= \frac{16 - (1 + 5 + 2\sqrt{5})}{16} \\
 &= \frac{10 - 2\sqrt{5}}{16} \\
 \cos 54^\circ &= \pm \frac{\sqrt{10 - 2\sqrt{5}}}{4}
 \end{aligned}$$

$\therefore \theta = 54^\circ$ lies in quad. I so $\cos 54^\circ > 0$

$$\cos 54^\circ = \frac{\sqrt{10 - 2\sqrt{5}}}{4}$$

Also $\cos 54^\circ = \cos(90^\circ - 36^\circ)$
 $= \sin 36^\circ$

$$\sin 36^\circ = \cos 54^\circ = \frac{\sqrt{10 - 2\sqrt{5}}}{4}$$

Next we show that

$$\cos 36^\circ \cos 72^\circ \cos 108^\circ \cos 144^\circ = \frac{1}{16}$$

$$\begin{aligned}
 \text{L.H.S.} &= \cos 36^\circ \cos 72^\circ \cos 108^\circ \cos 144^\circ \\
 &= \cos 36^\circ \cos 72^\circ \cos(180^\circ - 72^\circ) \cos(180^\circ - 36^\circ) \\
 &= \cos 36^\circ \cos 72^\circ (-\cos 72^\circ)(-\cos 36^\circ) \\
 &= \cos^2 36^\circ \cos^2 72^\circ \\
 &= \left(\frac{1 + \sqrt{5}}{4} \right)^2 \left(\frac{\sqrt{5} - 1}{4} \right)^2 \\
 &= \left(\frac{(\sqrt{5} + 1)(\sqrt{5} - 1)}{4 \times 4} \right)^2 \\
 &= \left(\frac{(\sqrt{5})^2 - (1)^2}{16} \right)^2 \\
 &= \left(\frac{5 - 1}{16} \right)^2 \\
 &= \left(\frac{4}{16} \right)^2 \\
 &= \left(\frac{1}{4} \right)^2 \\
 &= \frac{1}{16} = \text{R.H.S.}
 \end{aligned}$$

Sum, Difference and Product of Sines and Cosines:

$$2 \sin \alpha \cos \beta = \sin(\alpha + \beta) + \sin(\alpha - \beta)$$

$$2 \cos \alpha \sin \beta = \sin(\alpha + \beta) - \sin(\alpha - \beta)$$

$$2 \cos \alpha \cos \beta = \cos(\alpha + \beta) + \cos(\alpha - \beta)$$

$$-2 \sin \alpha \sin \beta = \cos(\alpha + \beta) - \cos(\alpha - \beta)$$

and

$$\sin P + \sin Q = 2 \sin\left(\frac{P+Q}{2}\right) \cos\left(\frac{P-Q}{2}\right)$$

$$\sin P - \sin Q = 2 \cos\left(\frac{P+Q}{2}\right) \sin\left(\frac{P-Q}{2}\right)$$

$$\cos P + \cos Q = 2 \cos\left(\frac{P+Q}{2}\right) \cos\left(\frac{P-Q}{2}\right)$$

$$\cos P - \cos Q = -2 \sin\left(\frac{P+Q}{2}\right) \sin\left(\frac{P-Q}{2}\right)$$

Exercise 10.4

Q.1 Express the following products as sum or differences:

(i) $2\sin 3\theta \cos \theta$

Solution:

We know that

$$\begin{aligned} &2\sin \alpha \cos \beta \\ &= \sin(\alpha + \beta) + \sin(\alpha - \beta) \end{aligned}$$

So

$$\begin{aligned} &2\sin 3\theta \cos \theta \\ &= \sin(3\theta + \theta) + \sin(3\theta - \theta) \\ &= \sin 4\theta + \sin 2\theta \end{aligned}$$

(ii) $2\cos 5\theta \sin 3\theta$

Solution:

We know that

$$\begin{aligned} &2\cos \alpha \sin \beta \\ &= \sin(\alpha + \beta) - \sin(\alpha - \beta) \end{aligned}$$

So

$$\begin{aligned} &2\cos 5\theta \sin 3\theta \\ &= \sin(5\theta + 3\theta) - \sin(5\theta - 3\theta) \\ &= \sin 8\theta - \sin 2\theta \end{aligned}$$

(iii) $\sin 5\theta \cos 2\theta$

Solution:

$$\begin{aligned} &\sin 5\theta \cos 2\theta \\ &= \frac{1}{2}(2\sin 5\theta \cos 2\theta) \\ &= \frac{1}{2}[\sin(5\theta + 2\theta) + \sin(5\theta - 2\theta)] \\ &= \frac{1}{2}[\sin 7\theta + \sin 3\theta] \end{aligned}$$

(iv) $2\sin 7\theta \sin 2\theta$

Solution:

$$\begin{aligned} &2\sin 7\theta \sin 2\theta \\ &= -(-2\sin 7\theta \sin 2\theta) \end{aligned}$$

$$= -[\cos(7\theta + 2\theta) - \cos(7\theta - 2\theta)]$$

$$= -[\cos 9\theta - \cos 5\theta]$$

$$= \cos 5\theta - \cos 9\theta$$

(v) $\cos(x + y)\sin(x - y)$

Solution:

$$\begin{aligned} &\cos(x + y)\sin(x - y) \\ &= \frac{1}{2}2\cos(x + y)\sin(x - y) \\ &= \frac{1}{2}[\sin(x + y + x - y) - \sin(x + y - x + y)] \\ &= \frac{1}{2}[\sin 2x - \sin 2y] \end{aligned}$$

(vi) $\cos(2x + 30^\circ)\cos(2x - 30^\circ)$

Solution:

$$\begin{aligned} &\cos(2x + 30^\circ)\cos(2x - 30^\circ) \\ &= \frac{1}{2}[2\cos(2x + 30^\circ)\cos(2x - 30^\circ)] \\ &= \frac{1}{2}\left[\cos(2x + 30^\circ + 2x - 30^\circ) + \cos(2x + 30^\circ - 2x + 30^\circ)\right] \\ &= \frac{1}{2}[\cos 4x + \cos 60^\circ] \end{aligned}$$

(vii) $\sin 12^\circ \sin 46^\circ$

Solution:

$$\begin{aligned} &\sin 12^\circ \sin 46^\circ \\ &= \frac{-1}{2}[-2\sin 46^\circ \sin 12^\circ] \\ &= \frac{-1}{2}[\cos(46^\circ + 12^\circ) - \cos(46^\circ - 12^\circ)] \\ &= \frac{-1}{2}[\cos 58^\circ - \cos 34^\circ] \\ &= \frac{1}{2}[\cos 34^\circ - \cos 58^\circ] \end{aligned}$$

(viii) $\sin(x + 45^\circ) \sin(x - 45^\circ)$

Solution:

$$\begin{aligned} & \sin(x + 45^\circ) \sin(x - 45^\circ) \\ &= \frac{-1}{2} [-2 \sin(x + 45^\circ) \sin(x - 45^\circ)] \\ &= \frac{-1}{2} \left[\cos(x + 45^\circ + x - 45^\circ) - \cos(x + 45^\circ - x + 45^\circ) \right] \\ &= \frac{-1}{2} [\cos 2x - \cos 90^\circ] \\ &= \frac{1}{2} [\cos 90^\circ - \cos 2x] \end{aligned}$$

Q.2 Express the following sums or differences as products;

(i) $\sin 5\theta + \sin 3\theta$

Solution:

$$\begin{aligned} & \sin 5\theta + \sin 3\theta \\ &= 2 \sin\left(\frac{5\theta + 3\theta}{2}\right) \cos\left(\frac{5\theta - 3\theta}{2}\right) \\ &= 2 \sin \frac{8\theta}{2} \cos \frac{2\theta}{2} \\ &= 2 \sin 4\theta \cos \theta \end{aligned}$$

(ii) $\sin 8\theta - \sin 4\theta$

Solution:

$$\begin{aligned} & \sin 8\theta - \sin 4\theta \\ &= 2 \cos\left(\frac{8\theta + 4\theta}{2}\right) \sin\left(\frac{8\theta - 4\theta}{2}\right) \\ &= 2 \cos \frac{12\theta}{2} \sin \frac{4\theta}{2} \\ &= 2 \cos 6\theta \sin 2\theta \end{aligned}$$

(iii) $\cos 6\theta + \cos 3\theta$

Solution:

$$\begin{aligned} & \cos 6\theta + \cos 3\theta \\ &= 2 \cos \frac{6\theta + 3\theta}{2} \cos \frac{6\theta - 3\theta}{2} \end{aligned}$$

$$= 2 \cos \frac{9\theta}{2} \cos \frac{3\theta}{2}$$

(iv) $\cos 7\theta - \cos \theta$

Solution:

$$\begin{aligned} & \cos 7\theta - \cos \theta \\ &= -2 \sin \frac{7\theta + \theta}{2} \sin \frac{7\theta - \theta}{2} \\ &= -2 \sin \frac{8\theta}{2} \sin \frac{6\theta}{2} \\ &= -2 \sin 4\theta \sin 3\theta \end{aligned}$$

(v) $\cos 12^\circ + \cos 48^\circ$

Solution:

$$\begin{aligned} & \cos 12^\circ + \cos 48^\circ \\ &= 2 \cos \frac{12^\circ + 48^\circ}{2} \cos \frac{12^\circ - 48^\circ}{2} \\ &= 2 \cos \frac{60^\circ}{2} \cos \frac{-36^\circ}{2} \\ &= 2 \cos 30^\circ \cos(-18^\circ) \\ &= 2 \cos 30^\circ \cos 18^\circ \end{aligned}$$

(vi) $\sin(x + 30^\circ) + \sin(x - 30^\circ)$

Solution:

$$\begin{aligned} & \sin(x + 30^\circ) + \sin(x - 30^\circ) \\ &= 2 \sin\left(\frac{x + 30^\circ + x - 30^\circ}{2}\right) \cos\left(\frac{x + 30^\circ - x + 30^\circ}{2}\right) \\ &= 2 \sin \frac{2x}{2} \cos \frac{60^\circ}{2} \\ &= 2 \sin x \cos 30^\circ \end{aligned}$$

Q.3 Prove the following identities :

(i) $\frac{\sin 3x - \sin x}{\cos x - \cos 3x} = \cot 2x$

Solution:

$$\begin{aligned} \text{L.H.S.} &= \frac{\sin 3x - \sin x}{\cos x - \cos 3x} \\ &= \frac{2 \cos\left(\frac{3x + x}{2}\right) \sin\left(\frac{3x - x}{2}\right)}{-2 \sin\left(\frac{x + 3x}{2}\right) \sin\left(\frac{x - 3x}{2}\right)} \end{aligned}$$

$$\begin{aligned}
 &= \frac{2 \cos \frac{4x}{2} \sin \frac{2x}{2}}{-2 \sin \frac{4x}{2} \sin \frac{-2x}{2}} \\
 &= \frac{2 \cos 2x \sin x}{-2 \sin 2x \sin(-x)} \\
 &= \frac{2 \cos 2x \sin x}{-2 \sin 2x \sin(-x)} \\
 &= \frac{2 \cos 2x \sin x}{+2 \sin 2x \sin x} \\
 &= \cot 2x = \text{R.H.S.}
 \end{aligned}$$

(ii) $\frac{\sin 8x + \sin 2x}{\cos 8x + \cos 2x} = \tan 5x$

Solution:

$$\begin{aligned}
 \text{L.H.S.} &= \frac{\sin 8x + \sin 2x}{\cos 8x + \cos 2x} \\
 &= \frac{2 \sin \frac{8x+2x}{2} \cos \frac{8x-2x}{2}}{2 \cos \frac{8x+2x}{2} \cos \frac{8x-2x}{2}} \\
 &= \frac{2 \sin \frac{10x}{2} \cos \frac{6x}{2}}{2 \cos \frac{10x}{2} \cos \frac{6x}{2}} \\
 &= \frac{2 \sin 5x \cos 3x}{2 \cos 5x \cos 3x} \\
 &= \tan 5x = \text{R.H.S.}
 \end{aligned}$$

(iii) $\frac{\sin \alpha - \sin \beta}{\sin \alpha + \sin \beta} = \tan \left(\frac{\alpha - \beta}{2} \right) \cot \left(\frac{\alpha + \beta}{2} \right)$

Solution:

$$\begin{aligned}
 \text{L.H.S.} &= \frac{\sin \alpha - \sin \beta}{\sin \alpha + \sin \beta} \\
 &= \frac{2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}}{2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}} \\
 &= \tan \frac{\alpha - \beta}{2} \cot \frac{\alpha + \beta}{2} = \text{R.H.S.}
 \end{aligned}$$

Q.4 Prove that:

(i) $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ = 0$

Solution:

$$\begin{aligned}
 \text{L.H.S.} &= \cos 20^\circ + \cos 100^\circ + \cos 140^\circ \\
 &= 2 \cos \frac{20^\circ + 100^\circ}{2} \cos \frac{20^\circ - 100^\circ}{2} + \cos 140^\circ \\
 &= 2 \cos \left(\frac{120^\circ}{2} \right) \cos \left(\frac{-80^\circ}{2} \right) + \cos 140^\circ \\
 &= 2 \cos 60^\circ \cos(-40^\circ) + \cos 140^\circ \\
 &= 2 \times \frac{1}{2} \cos 40^\circ + \cos 140^\circ \\
 &= \cos 40^\circ + \cos 140^\circ \\
 &= 2 \cos \left(\frac{40^\circ + 140^\circ}{2} \right) \cos \left(\frac{40^\circ - 140^\circ}{2} \right) \\
 &= 2 \cos \left(\frac{180^\circ}{2} \right) \cos \left(\frac{-100^\circ}{2} \right) \\
 &= 2 \cos 90^\circ \cos(-50^\circ) \\
 &= 2(0) \cos(50^\circ) \\
 &= 0 = \text{R.H.S.}
 \end{aligned}$$

(ii) $\sin \left(\frac{\pi}{4} - \theta \right) \sin \left(\frac{\pi}{4} + \theta \right) = \frac{1}{2} \cos 2\theta$

Solution:

$$\begin{aligned}
 \text{L.H.S.} &= \sin \left(\frac{\pi}{4} - \theta \right) \sin \left(\frac{\pi}{4} + \theta \right) \\
 &= -\frac{1}{2} \left[-2 \sin \left(\frac{\pi}{4} - \theta \right) \sin \left(\frac{\pi}{4} + \theta \right) \right] \\
 &= -\frac{1}{2} \left[\cos \left(\frac{\pi}{4} - \theta + \frac{\pi}{4} + \theta \right) - \cos \left(\frac{\pi}{4} - \theta - \frac{\pi}{4} - \theta \right) \right] \\
 &= -\frac{1}{2} \left[\cos \left(\frac{\pi}{2} \right) - \cos(-2\theta) \right] \\
 &= -\frac{1}{2} [0 - \cos 2\theta] \\
 &= \frac{1}{2} \cos 2\theta = \text{R.H.S.}
 \end{aligned}$$

$$(iii) \quad \frac{\sin\theta + \sin 3\theta + \sin 5\theta + \sin 7\theta}{\cos\theta + \cos 3\theta + \cos 5\theta + \cos 7\theta} = \tan 4\theta$$

Solution:

$$\begin{aligned} \text{L.H.S.} &= \frac{\sin\theta + \sin 3\theta + \sin 5\theta + \sin 7\theta}{\cos\theta + \cos 3\theta + \cos 5\theta + \cos 7\theta} \\ &= \frac{(\sin 7\theta + \sin \theta) + (\sin 5\theta + \sin 3\theta)}{(\cos 7\theta + \cos \theta) + (\cos 5\theta + \cos 3\theta)} \\ &= \frac{2\sin\left(\frac{7\theta + \theta}{2}\right)\cos\left(\frac{7\theta - \theta}{2}\right) + 2\sin\left(\frac{5\theta + 3\theta}{2}\right)\cos\left(\frac{5\theta - 3\theta}{2}\right)}{2\cos\left(\frac{7\theta + \theta}{2}\right)\cos\left(\frac{7\theta - \theta}{2}\right) + 2\cos\left(\frac{5\theta + 3\theta}{2}\right)\cos\left(\frac{5\theta - 3\theta}{2}\right)} \\ &= \frac{2\sin\left(\frac{8\theta}{2}\right)\cos\left(\frac{6\theta}{2}\right) + 2\sin\left(\frac{8\theta}{2}\right)\cos\left(\frac{2\theta}{2}\right)}{2\cos\left(\frac{8\theta}{2}\right)\cos\left(\frac{6\theta}{2}\right) + 2\cos\left(\frac{8\theta}{2}\right)\cos\left(\frac{2\theta}{2}\right)} \\ &= \frac{2\sin 4\theta \cos 3\theta + 2\sin 4\theta \cos \theta}{2\cos 4\theta \cos 3\theta + 2\cos 4\theta \cos \theta} \\ &= \frac{2\sin 4\theta [\cos 3\theta + \cos \theta]}{2\cos 4\theta [\cos 3\theta + \cos \theta]} \\ &= \frac{\sin 4\theta}{\cos 4\theta} \\ &= \tan 4\theta = \text{R.H.S.} \end{aligned}$$

Q.5 Prove that:

$$(i) \quad \cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16}$$

Solution:

$$\begin{aligned} \text{L.H.S.} &= \cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ \\ &= \frac{1}{2} \cos 80^\circ \cos 40^\circ \cos 20^\circ \\ &= \frac{1}{4} \{2 \cos 80^\circ \cos 40^\circ\} \cos 20^\circ \\ &= \frac{1}{4} \{\cos(80^\circ + 40^\circ) + \cos(80^\circ - 40^\circ)\} \cos 20^\circ \\ &= \frac{1}{4} \{\cos 120^\circ + \cos 40^\circ\} \cos 20^\circ \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4} \left\{ \frac{-1}{2} + \cos 40^\circ \right\} \cos 20^\circ \\
&= \frac{-1}{8} \cos 20^\circ + \frac{1}{4} \cos 40^\circ \cos 20^\circ \\
&= \frac{-1}{8} \cos 20^\circ + \frac{1}{8} (2 \cos 40^\circ \cos 20^\circ) \\
&= \frac{-1}{8} \cos 20^\circ + \frac{1}{8} [\cos(40^\circ + 20^\circ) + \cos(40^\circ - 20^\circ)] \\
&= \frac{-1}{8} \cos 20^\circ + \frac{1}{8} (\cos 60^\circ + \cos 20^\circ) \\
&= \frac{-1}{8} \cos 20^\circ + \frac{1}{8} \left(\frac{1}{2} + \cos 20^\circ \right) \\
&= \frac{-1}{8} \cos 20^\circ + \frac{1}{16} + \frac{1}{8} \cos 20^\circ \\
&= \frac{1}{16} = \text{R.H.S.}
\end{aligned}$$

(ii) $\sin \frac{\pi}{9} \sin \frac{2\pi}{9} \sin \frac{\pi}{3} \sin \frac{4\pi}{9} = \frac{3}{16}$

Solution:

$$\begin{aligned}
\text{L.H.S.} &= \sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ \\
&= \frac{\sqrt{3}}{2} [\sin 80^\circ \sin 40^\circ] \sin 20^\circ \\
&= -\frac{\sqrt{3}}{4} [-2 \sin 80^\circ \sin 40^\circ] \sin 20^\circ \\
&= \frac{-\sqrt{3}}{4} [\cos(80^\circ + 40^\circ) - \cos(80^\circ - 40^\circ)] \sin 20^\circ \\
&= \frac{-\sqrt{3}}{4} [\cos 120^\circ - \cos 40^\circ] \sin 20^\circ \\
&= \frac{-\sqrt{3}}{4} \left[\frac{-1}{2} - \cos 40^\circ \right] \sin 20^\circ \\
&= \frac{\sqrt{3}}{8} \sin 20^\circ + \frac{\sqrt{3}}{4} \cos 40^\circ \sin 20^\circ \\
&= \frac{\sqrt{3}}{8} \sin 20^\circ + \frac{\sqrt{3}}{8} (2 \cos 40^\circ \sin 20^\circ)
\end{aligned}$$

$$\begin{aligned}
&= \frac{\sqrt{3}}{8} \sin 20^\circ + \frac{\sqrt{3}}{8} [\sin(40^\circ + 20^\circ) - \sin(40^\circ - 20^\circ)] \\
&= \frac{\sqrt{3}}{8} \sin 20^\circ + \frac{\sqrt{3}}{8} (\sin 60^\circ - \sin 20^\circ) \\
&= \frac{\sqrt{3}}{8} \sin 20^\circ + \frac{\sqrt{3}}{8} \left(\frac{\sqrt{3}}{2} - \sin 20^\circ \right) \\
&= \frac{\sqrt{3}}{8} \sin 20^\circ + \frac{3}{16} - \frac{\sqrt{3}}{8} \sin 20^\circ \\
&= \frac{3}{16} = \text{R.H.S.}
\end{aligned}$$

(iii) $\sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ = \frac{1}{16}$

Solution:

$$\begin{aligned}
\text{L.H.S.} &= \sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ \\
&= \frac{1}{2} (\sin 70^\circ \sin 50^\circ) \sin 10^\circ \\
&= \frac{1}{-4} (-2 \sin 70^\circ \sin 50^\circ) \sin 10^\circ \\
&= \frac{1}{-4} [\cos(70^\circ + 50^\circ) - \cos(70^\circ - 50^\circ)] \sin 10^\circ \\
&= \frac{-1}{4} [\cos 120^\circ - \cos 20^\circ] \sin 10^\circ \\
&= \frac{-1}{4} \left[\frac{-1}{2} - \cos 20^\circ \right] \sin 10^\circ \\
&= \frac{1}{8} \sin 10^\circ + \frac{1}{4} \cos 20^\circ \sin 10^\circ \\
&= \frac{1}{8} \sin 10^\circ + \frac{1}{8} (2 \cos 20^\circ \sin 10^\circ) \\
&= \frac{1}{8} \sin 10^\circ + \frac{1}{8} [\sin(20^\circ + 10^\circ) - \sin(20^\circ - 10^\circ)] \\
&= \frac{1}{8} \sin 10^\circ + \frac{1}{8} (\sin 30^\circ - \sin 10^\circ) \\
&= \frac{1}{8} \sin 10^\circ + \frac{1}{8} \left(\frac{1}{2} - \sin 10^\circ \right) \\
&= \frac{1}{8} \sin 10^\circ + \frac{1}{16} - \frac{1}{8} \sin 10^\circ \\
&= \frac{1}{16} = \text{R.H.S.}
\end{aligned}$$